

CSE299

Graph Based ML codebase

(Group - 4)

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main.py model_config.py dataset_config.py X run.sh

```
codeP > dataset_config.py > ...
1 valid_datasets = {}
2     "node_classification": ["cora"],
3     "node_regression": ["karate_regression"],
4     "link_prediction": ["movielens"],
5     "edge_classification": ["cora_edge"],
6     "edge_regression": ["movielens_edge"],
7     "node_clustering": ["cora"],
8     "node_embedding": ["cora"],
9     "graph_classification": ["mutag"],
10    "graph_regression": ["synthetic_regression"],
11    "graph_matching": ["graph_matching"],
12    "graph_reconstruction": ["cora_reconstruction"],
13    "graph_generation": ["graph_generation"],
14 }
```

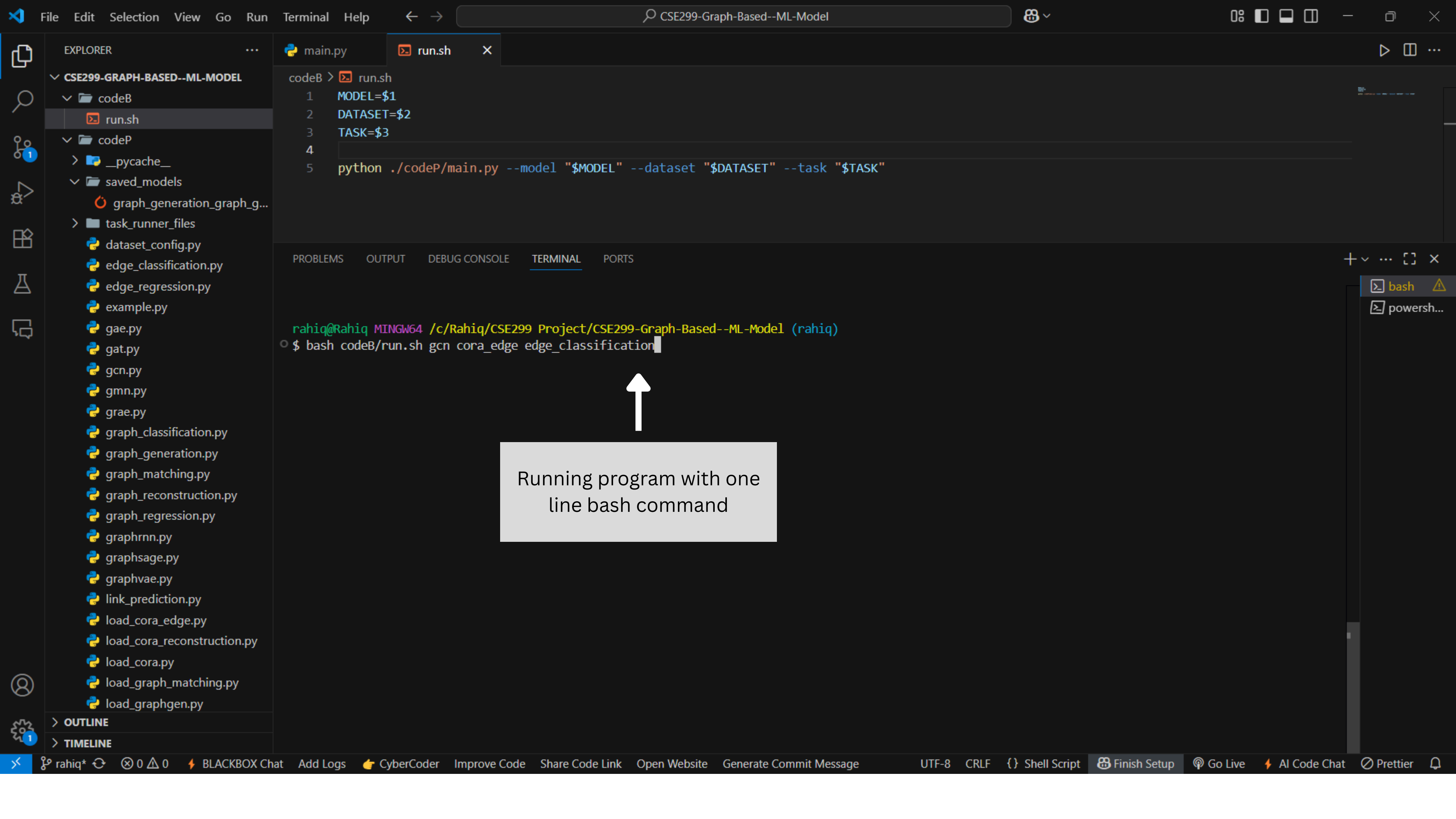


Valid datasets for
each task

All available models for
each task

```
# Models
available_models = {
    "gcn": {
        "name": "GCN",
        "module": {
            "link_prediction": link_prediction_gcn,
            "node_classification": node_classification_gcn,
            "node_regression": node_regression_gcn,
            "edge_classification": edge_classification_gcn,
            "edge_regression": edge_regression_gcn,
            "node_clustering": node_clustering_gcn,
            "node_embedding": node_embedding_gcn,
            "graph_classification": graph_classification_gcn,
            "graph_regression": graph_regression_gcn,
        },
    },
    "gat": {
        "name": "GAT",
        "module": {
            "link_prediction": link_prediction_gat,
            "node_classification": node_classification_gat,
            "node_regression": node_regression_gat,
            "edge_classification": edge_classification_gat,
            "edge_regression": edge_regression_gat,
            "node_clustering": node_clustering_gat,
            "node_embedding": node_embedding_gat,
            "graph_classification": graph_classification_gat,
            "graph_regression": graph_regression_gat,
        },
    },
    "graphsage": {
        "name": "GraphSAGE",
        "module": {
            "link_prediction": link_prediction_graphsage,
            "node_classification": node_classification_graphsage,
            "node_regression": node_regression_graphsage,
```

```
            "edge_classification": edge_classification_graphsage,
            "edge_regression": edge_regression_graphsage,
            "node_clustering": node_clustering_graphsage,
            "node_embedding": node_embedding_graphsage,
            "graph_classification": graph_classification_graphsage,
            "graph_regression": graph_regression_graphsage,
        },
    },
    "siamese_gcn": {
        "name": "Siamese GCN",
        "module": {
            "graph_matching": graph_matching_siamese_gcn,
        },
    },
    "gae": {
        "name": "GAE",
        "module": {
            "graph_reconstruction": graph_reconstruction_gae,
        },
    },
    "graphvae": {
        "name": "GraphVAE",
        "module": {
            "graph_generation": graph_generation_graphvae,
        },
    },
    "gmn": {
        "module": {
            "graph_matching": graph_matching_gmn,
        },
    },
    "graphrnn": {
        "module": {
            "graph_generation": graph_generation_graphrnn,
        },
    },
    "grae": {
        "module": {
            "graph_reconstruction": graph_reconstruction_grae,
```



Running program with one
line bash command

editSelectionViewGoRunTerminalHelp

CSE299-Graph-Based--ML-Model

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load_graphgen.py

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codeP > edge_classification.py > ...

6

7

8#GCN

9def edge_classification_gcn(x, edge_index, edge_labels, train_mask, test_mask, epochs=200):

10model = GCN(input_dim=x.size(1), hidden_dim=16, output_dim=16)

11classifier = torch.nn.Linear(16, 2)

12optimizer = torch.optim.Adam(list(model.parameters()) + list(classifier.parameters()), lr=0.01)

13

14for epoch in range(epochs):

15model.train()

16classifier.train()

17optimizer.zero_grad()

18

19node_embeddings = model(x, edge_index)

20src = edge_index[0]

21dst = edge_index[1]

22edge_emb = node_embeddings[src] * node_embeddings[dst]

23

24edge_logits = classifier(edge_emb)

25loss = F.cross_entropy(edge_logits[train_mask], edge_labels[train_mask])

26

27loss.backward()

28optimizer.step()

29

30if epoch % 20 == 0:

31model.eval()

32classifier.eval()

33with torch.no_grad():

34preds = edge_logits[test_mask].argmax(dim=1)

35acc = (preds == edge_labels[test_mask]).float().mean().item()

36print(f"[GCN-EdgeCls] Epoch {epoch} - Loss: {loss.item():.4f} - Test Acc: {acc:.4f}")

37

38return model, classifier

39

40

41#GAT

42def edge_classification_gat(x, edge_index, edge_labels, train_mask, test_mask, epochs=200):

Model training code for edge_classification

0000

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{ } Python

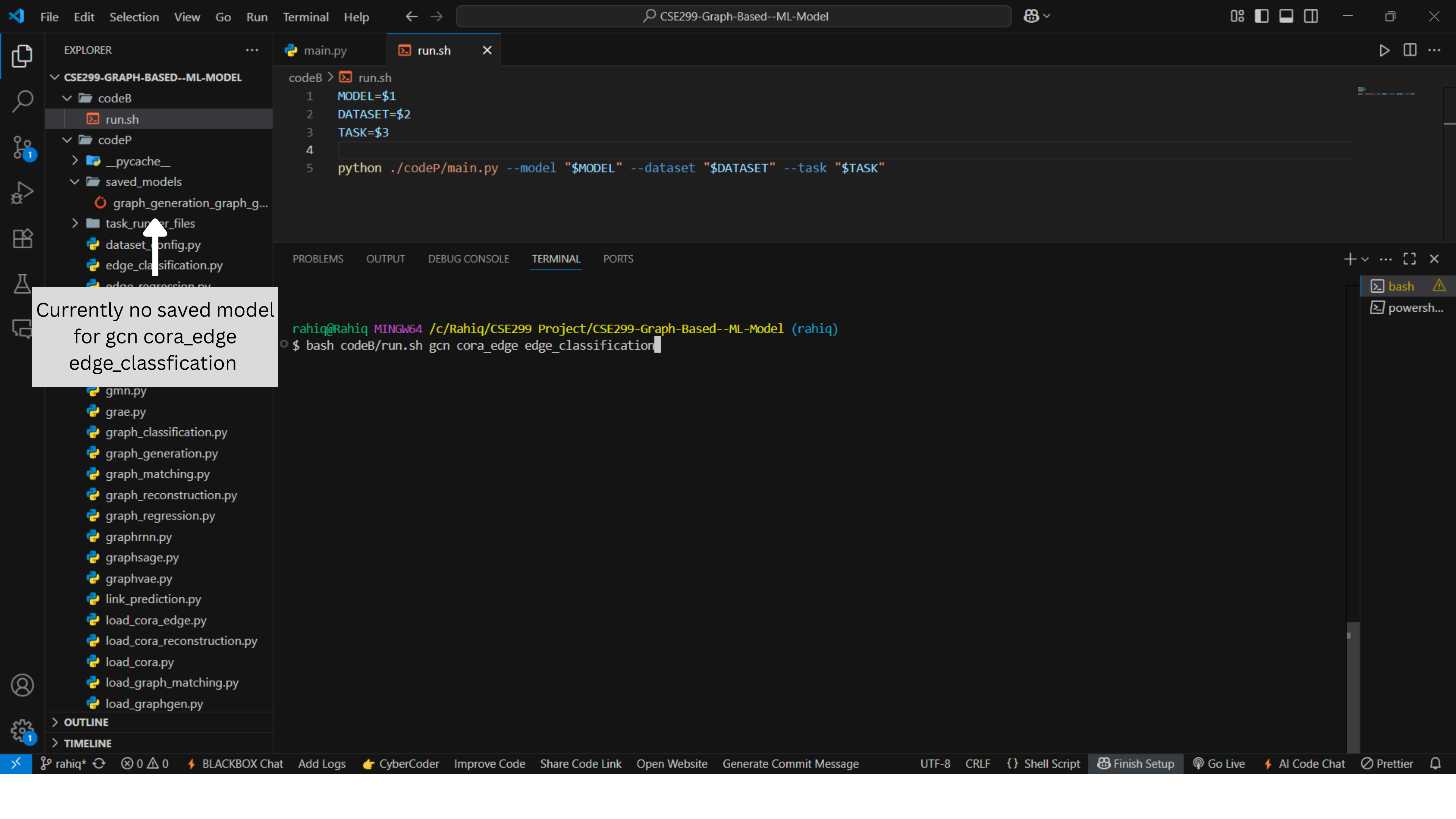
Finish Setup

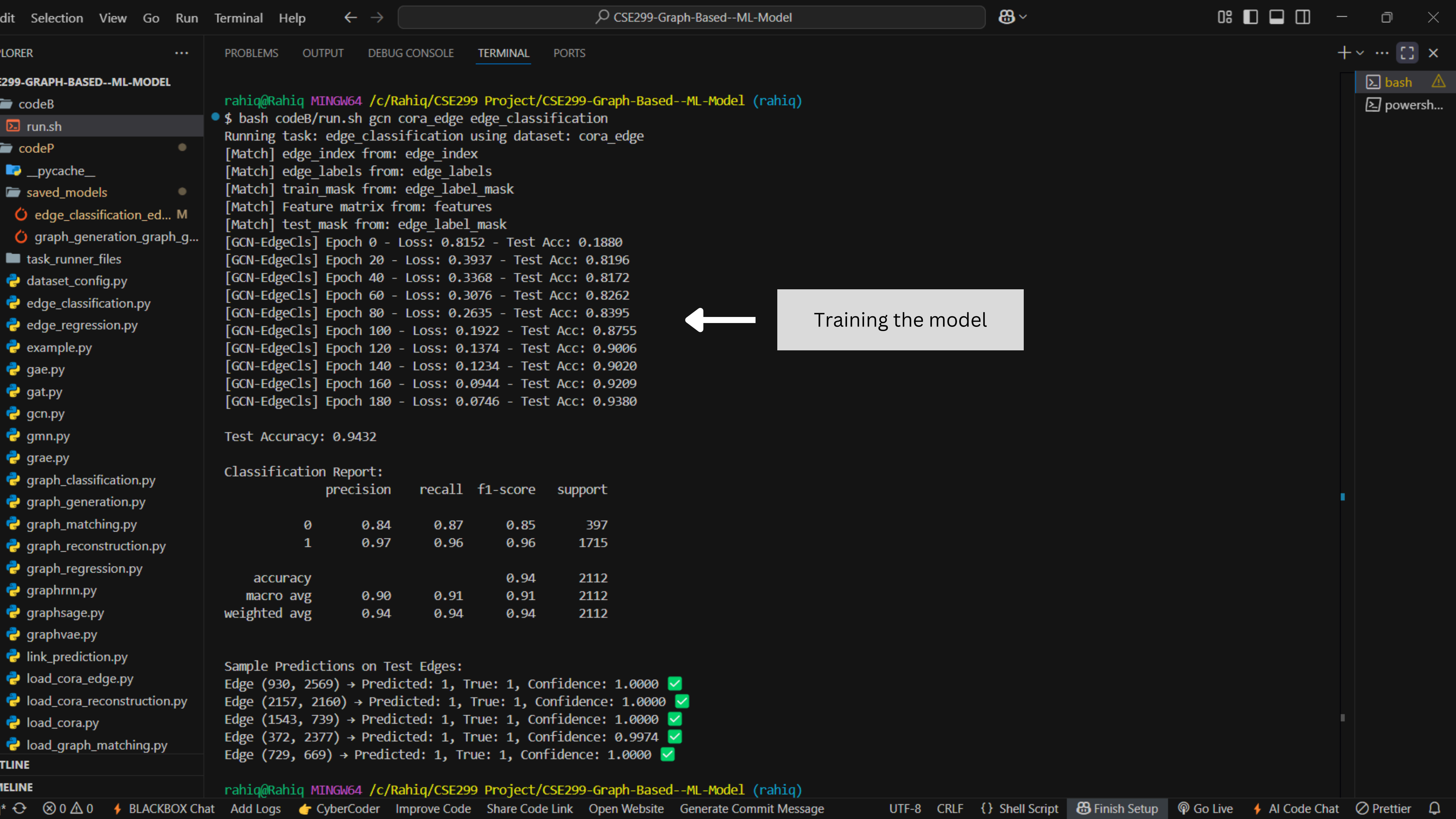
3.11.5 (venv)

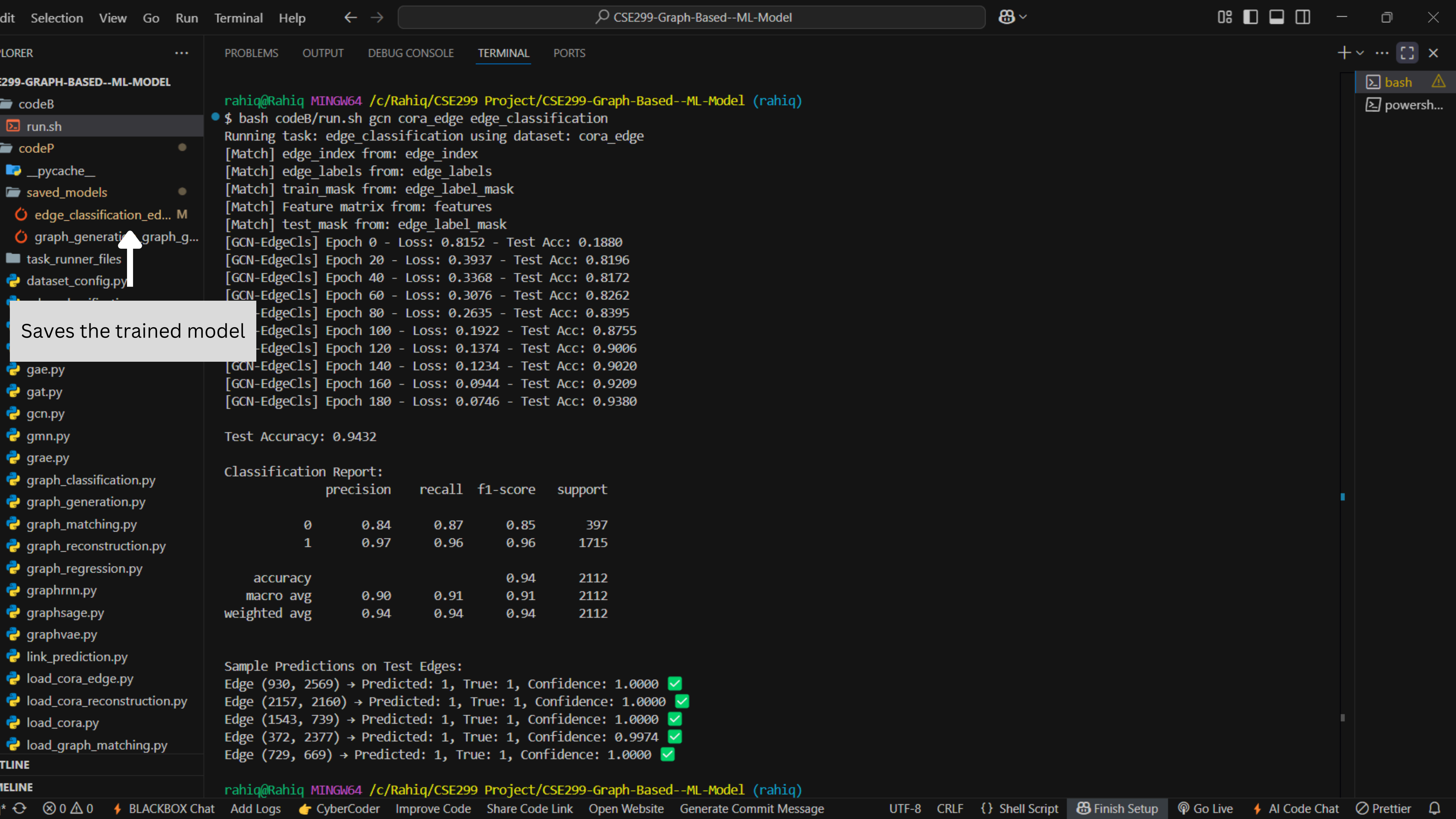
Go Live

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Saves the trained model

```
rahik@Rahik MINGW64 /c/Rahik/CSE299 Project/CSE299-Graph-Based--ML-Model (rahik)
$ bash codeB/run.sh gcn cora_edge edge_classification
Running task: edge_classification using dataset: cora_edge
[Match] edge_index from: edge_index
[Match] edge_labels from: edge_labels
[Match] train_mask from: edge_label_mask
[Match] Feature matrix from: features
[Match] test_mask from: edge_label_mask
[GCN-EdgeCls] Epoch 0 - Loss: 0.8152 - Test Acc: 0.1880
[GCN-EdgeCls] Epoch 20 - Loss: 0.3937 - Test Acc: 0.8196
[GCN-EdgeCls] Epoch 40 - Loss: 0.3368 - Test Acc: 0.8172
[GCN-EdgeCls] Epoch 60 - Loss: 0.3076 - Test Acc: 0.8262
[GCN-EdgeCls] Epoch 80 - Loss: 0.2635 - Test Acc: 0.8395
[GCN-EdgeCls] Epoch 100 - Loss: 0.1922 - Test Acc: 0.8755
[GCN-EdgeCls] Epoch 120 - Loss: 0.1374 - Test Acc: 0.9006
[GCN-EdgeCls] Epoch 140 - Loss: 0.1234 - Test Acc: 0.9020
[GCN-EdgeCls] Epoch 160 - Loss: 0.0944 - Test Acc: 0.9209
[GCN-EdgeCls] Epoch 180 - Loss: 0.0746 - Test Acc: 0.9380

Test Accuracy: 0.9432

Classification Report:
              precision    recall  f1-score   support

         0       0.84      0.87      0.85        397
         1       0.97      0.96      0.96       1715

   accuracy                0.94        2112
  macro avg       0.90      0.91      0.91        2112
 weighted avg     0.94      0.94      0.94        2112

Sample Predictions on Test Edges:
Edge (930, 2569) -> Predicted: 1, True: 1, Confidence: 1.0000 ✓
Edge (2157, 2160) -> Predicted: 1, True: 1, Confidence: 1.0000 ✓
Edge (1543, 739) -> Predicted: 1, True: 1, Confidence: 1.0000 ✓
Edge (372, 2377) -> Predicted: 1, True: 1, Confidence: 0.9974 ✓
Edge (729, 669) -> Predicted: 1, True: 1, Confidence: 1.0000 ✓

rahik@Rahik MINGW64 /c/Rahik/CSE299 Project/CSE299-Graph-Based--ML-Model (rahik)
```

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weighted avg

0.94

0.94

0.94

2112

Sample Predictions on Test Edges:

Edge (930, 2569) → Predicted: 1, True: 1, Confidence: 1.0000 ✓

Edge (2157, 2160) → Predicted: 1, True: 1, Confidence: 1.0000 ✓

Edge (1543, 739) → Predicted: 1, True: 1, Confidence: 1.0000 ✓

Edge (372, 2377) → Predicted: 1, True: 1, Confidence: 0.9974 ✓

Edge (729, 669) → Predicted: 1, True: 1, Confidence: 1.0000 ✓

rahiiq@Rahiiq MINGW64 /c/Rahiiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiiq)

\$ bash codeB/run.sh gcn cora_edge edge_classification

Running task: edge_classification using dataset: cora_edge

[Match] edge_index from: edge_index

[Match] edge_labels from: edge_labels

[Match] train_mask from: edge_label_mask

[Match] Feature matrix from: features

[Match] test_mask from: edge_label_mask

Test Accuracy: 0.9432

Classification Report:

	precision	recall	f1-score	support
0	0.84	0.87	0.85	397
1	0.97	0.96	0.96	1715
accuracy			0.94	2112
macro avg	0.90	0.91	0.91	2112
weighted avg	0.94	0.94	0.94	2112

Sample Predictions on Test Edges:

Edge (841, 2034) → Predicted: 0, True: 0, Confidence: 0.9905 ✓

Edge (680, 2199) → Predicted: 1, True: 1, Confidence: 1.0000 ✓

Edge (734, 2202) → Predicted: 1, True: 1, Confidence: 1.0000 ✓

Edge (1971, 1702) → Predicted: 1, True: 1, Confidence: 1.0000 ✓

Edge (1317, 1292) → Predicted: 1, True: 1, Confidence: 1.0000 ✓

rahiiq@Rahiiq MINGW64 /c/Rahiiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiiq)

\$

Running the same task again

Using the saved model to do the task

bash

powersh...

0 0

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{ } Shell Script

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eP > loader_config.py > ...

```

1 import os
2 import json
3 import torch
4 import pickle
5 import pandas as pd
6 import numpy as np
7 from torch_geometric.data import Data
8 from torch_geometric.data.data import DataEdgeAttr
9
10 torch.serialization.add_safe_globals([Data, DataEdgeAttr])
11
12 def try_loaders(path):
13     loaders = [
14         lambda p: torch.load(p, weights_only=False),
15         lambda p: pickle.load(open(p, 'rb')),
16         lambda p: np.load(p, allow_pickle=True),
17         lambda p: pd.read_csv(p, header=None).values,
18         lambda p: json.load(open(p, 'r', encoding='utf-8')),
19         lambda p: [json.loads(line) for line in open(p, 'r', encoding='utf-8')],
20         lambda p: open(p, 'r', encoding='utf-8').read(),
21     ]
22     for loader in loaders:
23         try:
24             return loader(path)
25         except Exception:
26             continue
27     print(f"[Warning] Could not load file: {path}")
28     return None
29
30 def load_file(path):
31     try:
32         return try_loaders(path)
33     except Exception as e:
34         print(f"[Error] Exception while loading {path}: {e}")
35         return None
  
```



Supports different file
formats

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graphvae.py

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load_cora_reconstruction.py

load_cora.py

codeB > run.sh

1 MODEI=\$1

codeP > dataset_config.py > ...

1 valid_datasets = {

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3. Item 1268 → Score: 0.7494

4. Item 1930 → Score: 0.7480

5. Item 1265 → Score: 0.7467

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh gcn movielens link_prediction

Running task: link_prediction using dataset: movielens

[Match] Train edge_index from: edge_index

[Match] Feature matrix from: features

[Match] Test edge_index from: edge_index

Loading saved model from: codeP/saved_models/link_prediction_link_prediction_gcn_movielens.pt

Sample Link Predictions:

Top recommendations for user 601:

1. Item 1276 → Score: 0.7900

2. Item 1265 → Score: 0.7550

3. Item 1930 → Score: 0.7467

4. Item 1632 → Score: 0.7448

5. Item 1268 → Score: 0.7378

Top recommendations for user 156:

1. Item 1265 → Score: 0.8195

2. Item 1187 → Score: 0.8006

3. Item 1285 → Score: 0.7914

4. Item 1930 → Score: 0.7899

5. Item 1243 → Score: 0.7856

Top recommendations for user 641:

1. Item 1229 → Score: 0.9935

2. Item 1951 → Score: 0.9892

3. Item 1035 → Score: 0.9888

4. Item 1876 → Score: 0.9864

5. Item 1659 → Score: 0.9857

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$

bash

powersh...

powersh...

0 0

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rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh gcn cora node_classification

Running task: node_classification using dataset: cora

[Match] edge_index from: edge_index

[Match] Feature matrix from: features

[Match] labels from: labels

[Match] train_mask from: train_mask

[Match] test_mask from: test_mask

Loading saved model from: codeP/saved_models/node_classification_gcn cora.pt

Evaluation on Test Set:

Accuracy: 0.7800

Classification Report:

	precision	recall	f1-score	support
0	0.66	0.75	0.70	130
1	0.71	0.86	0.78	91
2	0.79	0.90	0.84	144
3	0.94	0.68	0.79	319
4	0.75	0.85	0.79	149
5	0.77	0.77	0.77	103
6	0.69	0.81	0.75	64
accuracy			0.78	1000
macro avg	0.76	0.80	0.77	1000
weighted avg	0.80	0.78	0.78	1000

Sample Node Predictions (Test Set):

Node 1972 → Predicted: 6, Actual: 6 ✓

Node 2193 → Predicted: 0, Actual: 0 ✓

Node 2345 → Predicted: 2, Actual: 3 ✗

Node 1799 → Predicted: 1, Actual: 3 ✗

Node 2574 → Predicted: 3, Actual: 3 ✓

bash

powersh...

powersh...

Figure 1

t-SNE of Node Embeddings (Test Set)

Class 0

Class 1

Class 2

Class 3

Class 4

Class 5

Class 6

TSNE-2

TSNE-1

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rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh gcn karate_regression node_regression

Running task: node_regression using dataset: karate_regression

[Match] edge_index from: edge_index

[Match] Feature matrix from: features

[Match] targets from: targets

[Match] train_mask from: train_mask

[Match] test_mask from: test_mask

No saved model found. Training from scratch...

[GCN-NodeReg] Epoch 0 - Loss: 13.7964 - Test MSE: 4.1609

[GCN-NodeReg] Epoch 20 - Loss: 4.6479 - Test MSE: 0.3594

[GCN-NodeReg] Epoch 40 - Loss: 2.5046 - Test MSE: 2.1338

[GCN-NodeReg] Epoch 60 - Loss: 1.9763 - Test MSE: 0.8149

[GCN-NodeReg] Epoch 80 - Loss: 1.7009 - Test MSE: 1.1180

[GCN-NodeReg] Epoch 100 - Loss: 1.4523 - Test MSE: 0.9659

[GCN-NodeReg] Epoch 120 - Loss: 1.2036 - Test MSE: 0.9511

[GCN-NodeReg] Epoch 140 - Loss: 0.9628 - Test MSE: 0.9056

[GCN-NodeReg] Epoch 160 - Loss: 0.7501 - Test MSE: 0.8788

[GCN-NodeReg] Epoch 180 - Loss: 0.5725 - Test MSE: 0.8996

Evaluation on Test Set:

Mean Squared Error (MSE): 0.9407

Mean Absolute Error (MAE): 0.8344

R² Score: -1.4670

Sample Node Regression Results (Test Set):

Node 30 → Predicted: 3.2377 | Actual: 2.3529 | Absolute Error: 0.8848

Node 24 → Predicted: 2.9793 | Actual: 1.7647 | Absolute Error: 1.2146

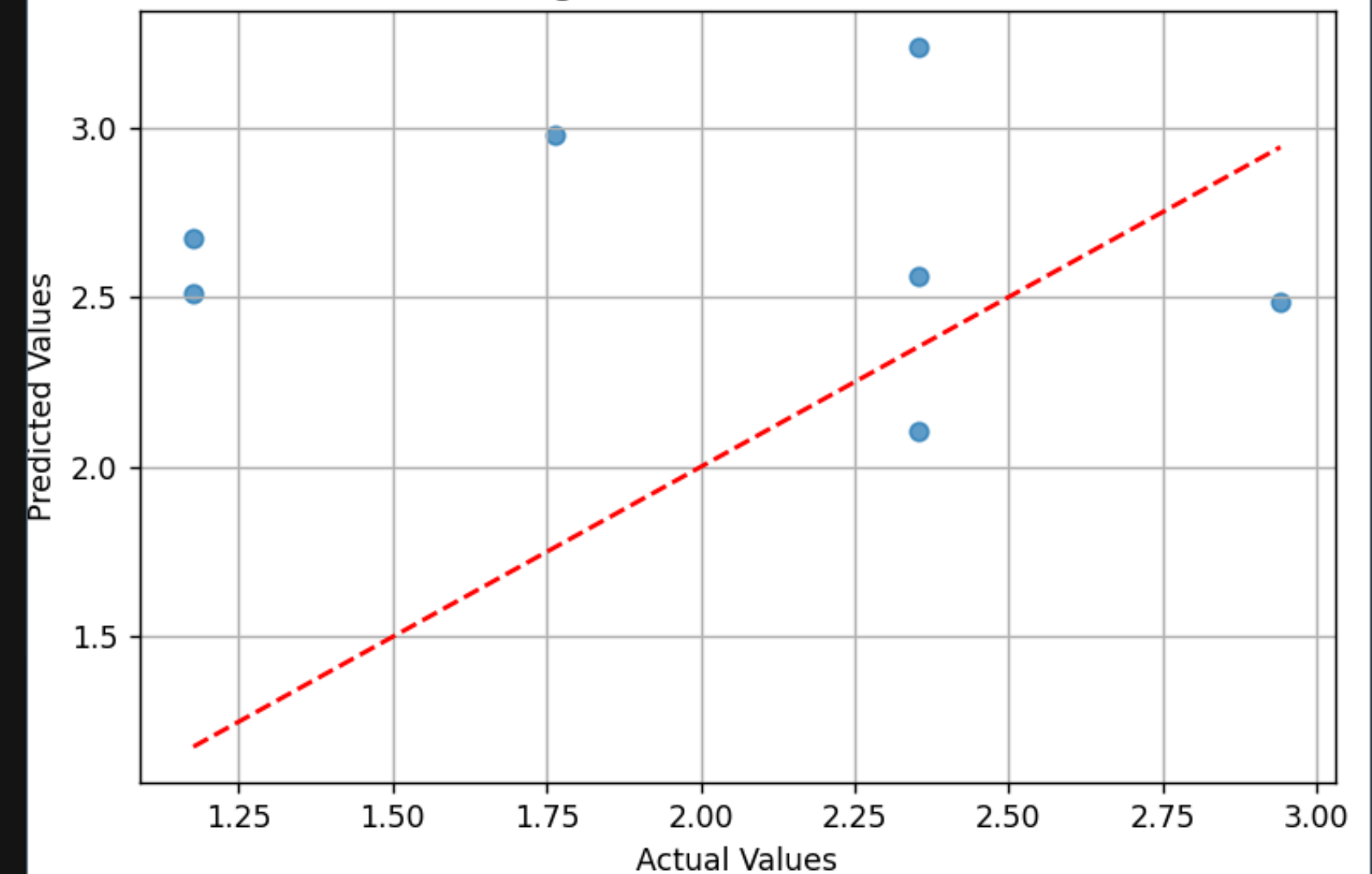
Node 5 → Predicted: 2.1044 | Actual: 2.3529 | Absolute Error: 0.2486

Node 22 → Predicted: 2.5104 | Actual: 1.1765 | Absolute Error: 1.3339

Node 27 → Predicted: 2.5598 | Actual: 2.3529 | Absolute Error: 0.2069

Figure 1

Node Regression: Predicted vs Actual



Navigation icons: Home, Previous, Next, Full Screen, Search, List, Save.

(x, y) = (1.920, 3.190)

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rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)

```
$ bash codeB/run.sh gcn karate_regression node_regression  
Running task: node_regression using dataset: karate_regression
```

```
[Match] edge_index from: edge_index
```

```
[Match] Feature matrix from: features
```

```
[Match] targets from: targets
```

```
[Match] train_mask from: train_mask
```

```
[Match] test_mask from: test_mask
```

```
No saved model found. Training from scratch...
```

```
[GCN-NodeReg] Epoch 0 - Loss: 13.7964 - Test MSE: 4.1609
```

```
[GCN-NodeReg] Epoch 20 - Loss: 4.6479 - Test MSE: 0.3594
```

```
[GCN-NodeReg] Epoch 40 - Loss: 2.5046 - Test MSE: 2.1338
```

```
[GCN-NodeReg] Epoch 60 - Loss: 1.9763 - Test MSE: 0.8149
```

```
[GCN-NodeReg] Epoch 80 - Loss: 1.7009 - Test MSE: 1.1180
```

```
[GCN-NodeReg] Epoch 100 - Loss: 1.4523 - Test MSE: 0.9659
```

```
[GCN-NodeReg] Epoch 120 - Loss: 1.2036 - Test MSE: 0.9511
```

```
[GCN-NodeReg] Epoch 140 - Loss: 0.9628 - Test MSE: 0.9056
```

```
[GCN-NodeReg] Epoch 160 - Loss: 0.7501 - Test MSE: 0.8788
```

```
[GCN-NodeReg] Epoch 180 - Loss: 0.5725 - Test MSE: 0.8996
```

```
Evaluation on Test Set:
```

```
Mean Squared Error (MSE): 0.9407
```

```
Mean Absolute Error (MAE): 0.8344
```

```
R2 Score: -1.4670
```

```
Sample Node Regression Results (Test Set):
```

```
Node 30 → Predicted: 3.2377 | Actual: 2.3529 | Absolute Error: 0.8848
```

```
Node 24 → Predicted: 2.9793 | Actual: 1.7647 | Absolute Error: 1.2146
```

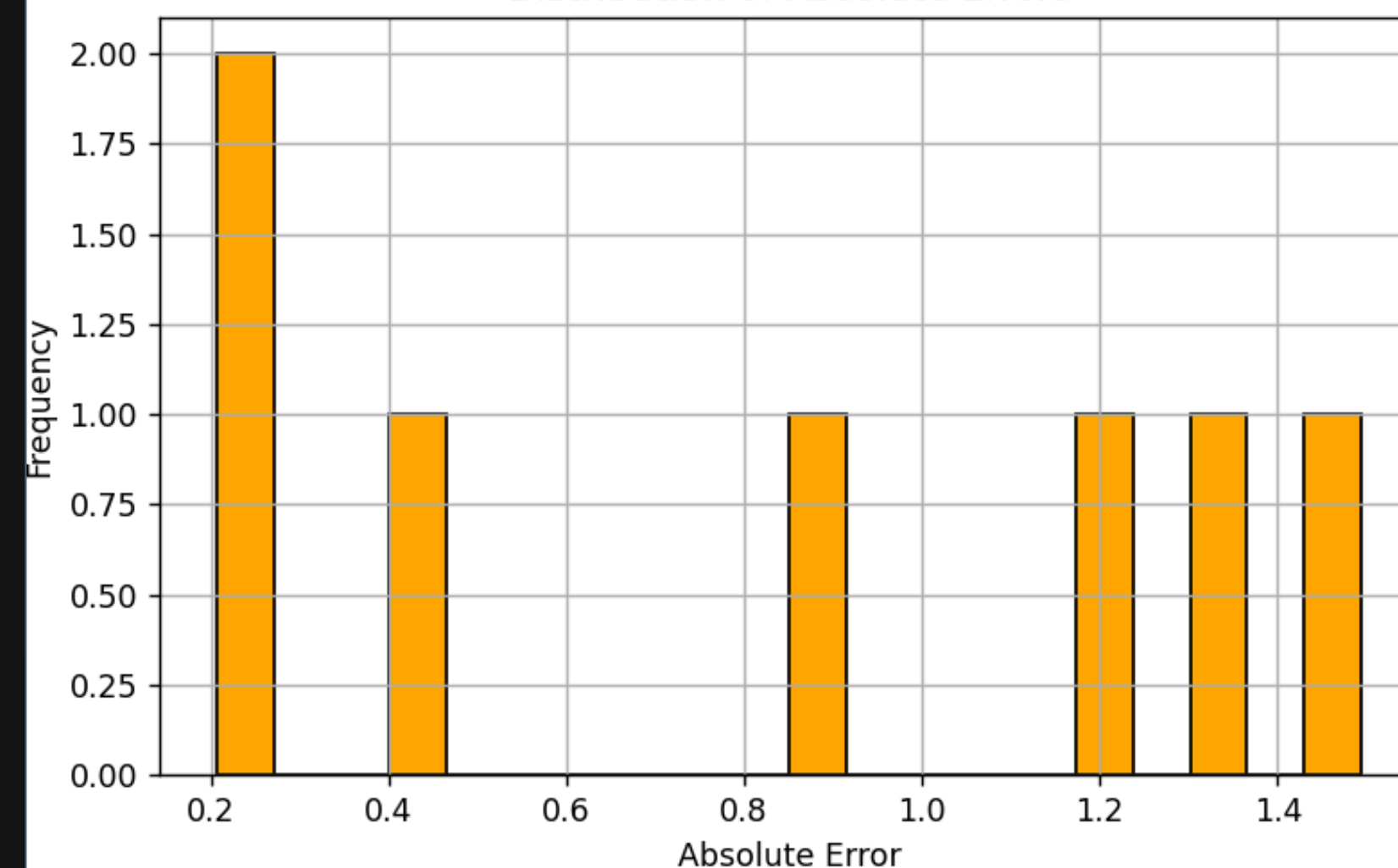
```
Node 5 → Predicted: 2.1044 | Actual: 2.3529 | Absolute Error: 0.2486
```

```
Node 22 → Predicted: 2.5104 | Actual: 1.1765 | Absolute Error: 1.3339
```

```
Node 27 → Predicted: 2.5598 | Actual: 2.3529 | Absolute Error: 0.2069
```

Figure 1

Distribution of Absolute Errors



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graphrnn.py
graphsage.py
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
R² Score: -1.4670
Sample Node Regression Results (Test Set):
Node 30 → Predicted: 3.2377 | Actual: 2.3529 | Absolute Error: 0.8848
Node 24 → Predicted: 2.9793 | Actual: 1.7647 | Absolute Error: 1.2146
Node 5 → Predicted: 2.1044 | Actual: 2.3529 | Absolute Error: 0.2486
Node 22 → Predicted: 2.5104 | Actual: 1.1765 | Absolute Error: 1.3339
Node 27 → Predicted: 2.5598 | Actual: 2.3529 | Absolute Error: 0.2069
rahii@Rahii MINGW64 /c/Rahii/CSE299 Project/CSE299-Graph-Based--ML-Model (rahii)
\$ bash codeB/run.sh gcn movielens_edge edge_regression
Running task: edge_regression using dataset: movielens_edge
[Match] edge_index from: edge_index
[Match] edge_ratings from: edge_ratings
[Match] train_mask from: edge_rating_mask
[Match] Feature matrix from: features
[Match] test_mask from: edge_rating_mask
[GCN-EdgeReg] Epoch 0 - Loss: 12.9105 - Test MSE: 12.8656
[GCN-EdgeReg] Epoch 20 - Loss: 4.0433 - Test MSE: 4.0406
[GCN-EdgeReg] Epoch 40 - Loss: 2.4905 - Test MSE: 2.4919
[GCN-EdgeReg] Epoch 60 - Loss: 1.9981 - Test MSE: 2.0005
[GCN-EdgeReg] Epoch 80 - Loss: 1.6525 - Test MSE: 1.6552
[GCN-EdgeReg] Epoch 100 - Loss: 1.3328 - Test MSE: 1.3329
[GCN-EdgeReg] Epoch 120 - Loss: 1.2271 - Test MSE: 1.2258
[GCN-EdgeReg] Epoch 140 - Loss: 1.1360 - Test MSE: 1.1368
[GCN-EdgeReg] Epoch 160 - Loss: 1.0577 - Test MSE: 1.0585
[GCN-EdgeReg] Epoch 180 - Loss: 1.0267 - Test MSE: 1.0270
Mean Squared Error (MSE): 1.0074
R² Score: 0.2032
Sample Predictions:
Edge (879, 1024) → Predicted: 3.79, Actual: 3.00, Error: 0.79
Edge (200, 1228) → Predicted: 3.57, Actual: 2.00, Error: 1.57
Edge (1677, 362) → Predicted: 3.60, Actual: 3.00, Error: 0.60
Edge (278, 1572) → Predicted: 3.36, Actual: 4.00, Error: 0.64
Edge (1265, 833) → Predicted: 3.02, Actual: 2.00, Error: 1.02
rahii@Rahii MINGW64 /c/Rahii/CSE299 Project/CSE299-Graph-Based--ML-Model (rahii)
\$

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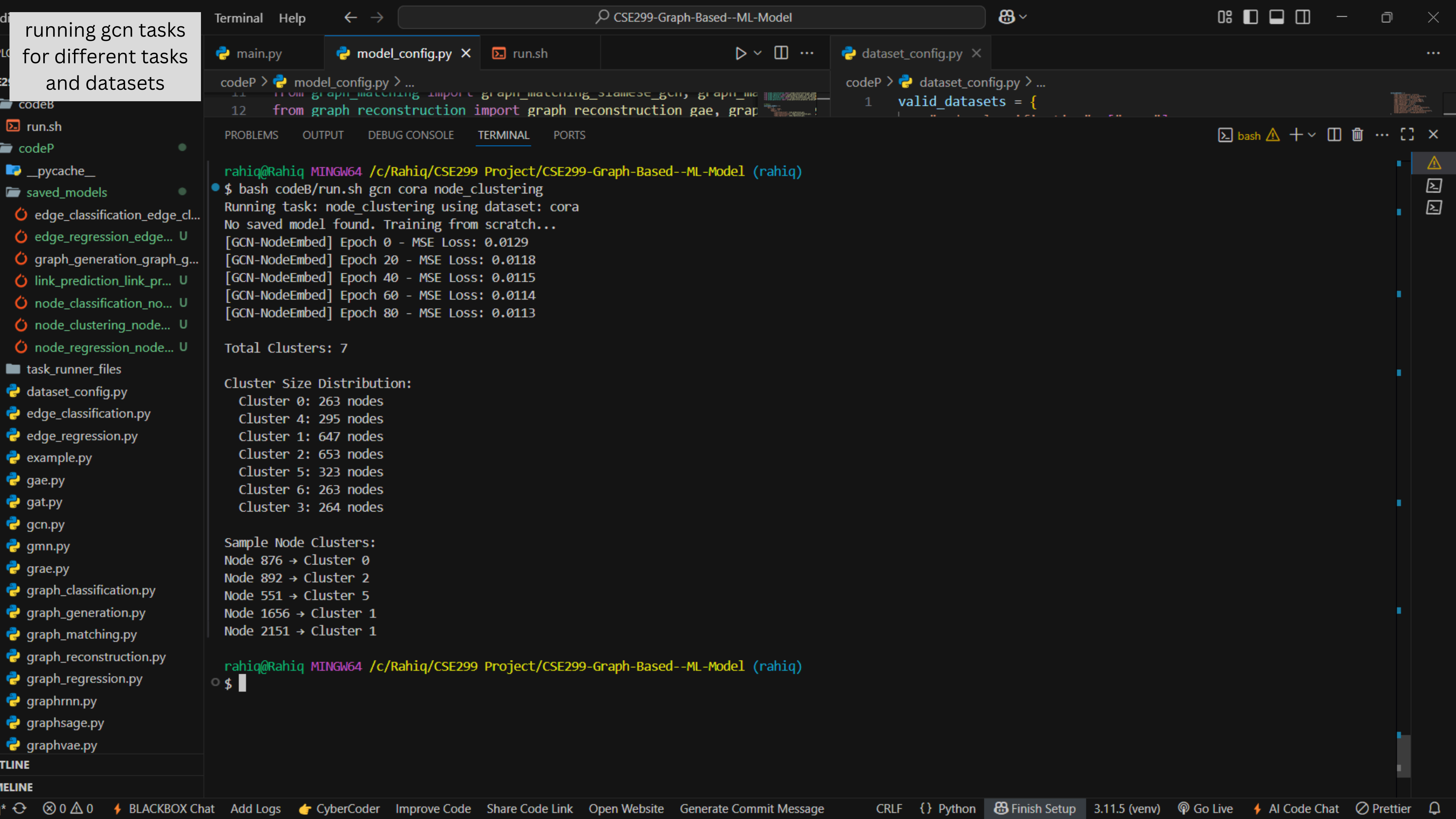
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CSE299-Graph-Based--ML-Model

main.py model_config.py x run.sh dataset_config.py x

codeP > model_config.py > ...
11 from graph_matching import graph_matching_siamese_gcn, graph_ma
12 from graph_reconstruction import graph_reconstruction_gae, grap

codeP > dataset_config.py > ...
1 valid_datasets = {

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

bash

rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)
\$ bash codeB/run.sh gcn cora node_clustering
Running task: node_clustering using dataset: cora
No saved model found. Training from scratch...
[GCN-NodeEmbed] Epoch 0 - MSE Loss: 0.0129
[GCN-NodeEmbed] Epoch 20 - MSE Loss: 0.0118
[GCN-NodeEmbed] Epoch 40 - MSE Loss: 0.0115
[GCN-NodeEmbed] Epoch 60 - MSE Loss: 0.0114
[GCN-NodeEmbed] Epoch 80 - MSE Loss: 0.0113

Total Clusters: 7

Cluster Size Distribution:
Cluster 0: 263 nodes
Cluster 4: 295 nodes
Cluster 1: 647 nodes
Cluster 2: 653 nodes
Cluster 5: 323 nodes
Cluster 6: 263 nodes
Cluster 3: 264 nodes

Sample Node Clusters:
Node 876 -> Cluster 0
Node 892 -> Cluster 2
Node 551 -> Cluster 5
Node 1656 -> Cluster 1
Node 2151 -> Cluster 1

rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)
\$

task_runner_files

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edge_regression.py

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gcn.py

gm n.py

grae.py

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graph_matching.py

graph_reconstruction.py

graph_regression.py

graphrnn.py

graphsage.py

graphvae.py

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node_classification_no...

node_clustering_node...

node_embedding_nod...

node_regression_node...

task_runner_files

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edge_classification.py

edge_regression.py

example.py

gae.py

gat.py

gcn.py

gmn.py

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graph_matching.py

graph_reconstruction.py

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graphsage.py

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main.py model_config.py X run.sh dataset_config.py X

codeP > model_config.py > ...
11 from graph_matching import graph_matching_siamese_gcn, graph_ma
12 from graph_reconstruction import graph_reconstruction_gae, gra
codeP > dataset_config.py > ...
1 valid_datasets = {

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rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)
\$ bash codeB/run.sh gcn cora node_embedding
Running task: node_embedding using dataset: cora
No saved model found. Training from scratch...
[GCN-NodeEmbed] Epoch 0 - MSE Loss: 0.0130
[GCN-NodeEmbed] Epoch 20 - MSE Loss: 0.0118
[GCN-NodeEmbed] Epoch 40 - MSE Loss: 0.0116
[GCN-NodeEmbed] Epoch 60 - MSE Loss: 0.0114
[GCN-NodeEmbed] Epoch 80 - MSE Loss: 0.0114

Sample Node Embeddings:
Node 0 -> Embedding: [0.0024, 0.0217, 0.0256, -0.0004, 0.0297, ...]
Node 1 -> Embedding: [-0.0051, -0.0124, -0.0063, 0.0396, -0.0502, ...]
Node 2 -> Embedding: [-0.0030, -0.0019, 0.0164, 0.0368, -0.0274, ...]
Node 3 -> Embedding: [-0.0009, -0.0048, 0.0119, 0.0371, 0.0227, ...]
Node 4 -> Embedding: [0.0073, 0.0125, 0.0227, 0.0262, 0.0915, ...]

Embedding Statistics:
Mean: 0.0127
Std Dev: 0.0340
Min: -0.4595
Max: 2.5026

Cosine Similarity Between Sample Nodes:
Node 0 vs Node 1: 0.5154
Node 0 vs Node 2: 0.6135
Node 0 vs Node 3: 0.2785
Node 0 vs Node 4: 0.4364
Node 1 vs Node 2: 0.9271
Node 1 vs Node 3: 0.7855
Node 1 vs Node 4: 0.6091
Node 2 vs Node 3: 0.8257
Node 2 vs Node 4: 0.5491
Node 3 vs Node 4: 0.5920

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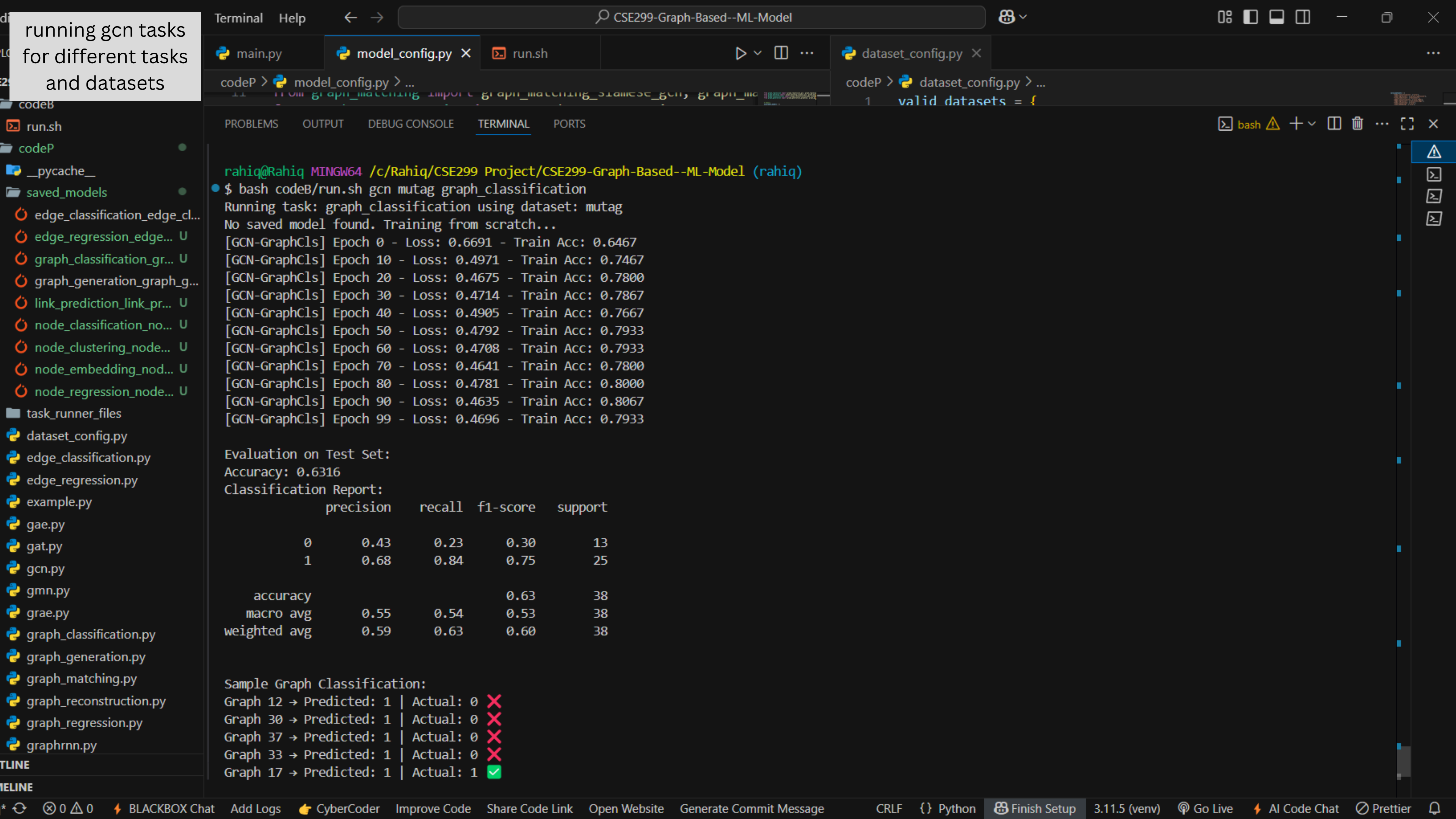
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rahilq@Rahilq MINGW64 /c/Rahilq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahilq)

```
$ bash codeB/run.sh gcn synthetic_regression graph_regression
Running task: graph_regression using dataset: synthetic_regression
```

No saved model found. Training from scratch...

[GCN-GraphReg] Epoch 0 - Train MSE: 97.183635

[GCN-GraphReg] Epoch 10 - Train MSE: 5.703851

[GCN-GraphReg] Epoch 20 - Train MSE: 4.431348

[GCN-GraphReg] Epoch 30 - Train MSE: 5.535823

[GCN-GraphReg] Epoch 40 - Train MSE: 3.090857

[GCN-GraphReg] Epoch 50 - Train MSE: 3.097077

[GCN-GraphReg] Epoch 60 - Train MSE: 2.237914

[GCN-GraphReg] Epoch 70 - Train MSE: 1.971153

[GCN-GraphReg] Epoch 80 - Train MSE: 1.883828

[GCN-GraphReg] Epoch 90 - Train MSE: 2.248514

[GCN-GraphReg] Epoch 99 - Train MSE: 1.946905

Evaluation on Test Set:

MSE: 2.3075

MAE: 1.1903

R² Score: 0.3953

Sample Graph Regression:

Graph 17 → Predicted: 24.0982 | True: 23.8365 | Abs Error: 0.2618

Graph 13 → Predicted: 25.1579 | True: 27.5053 | Abs Error: 2.3475

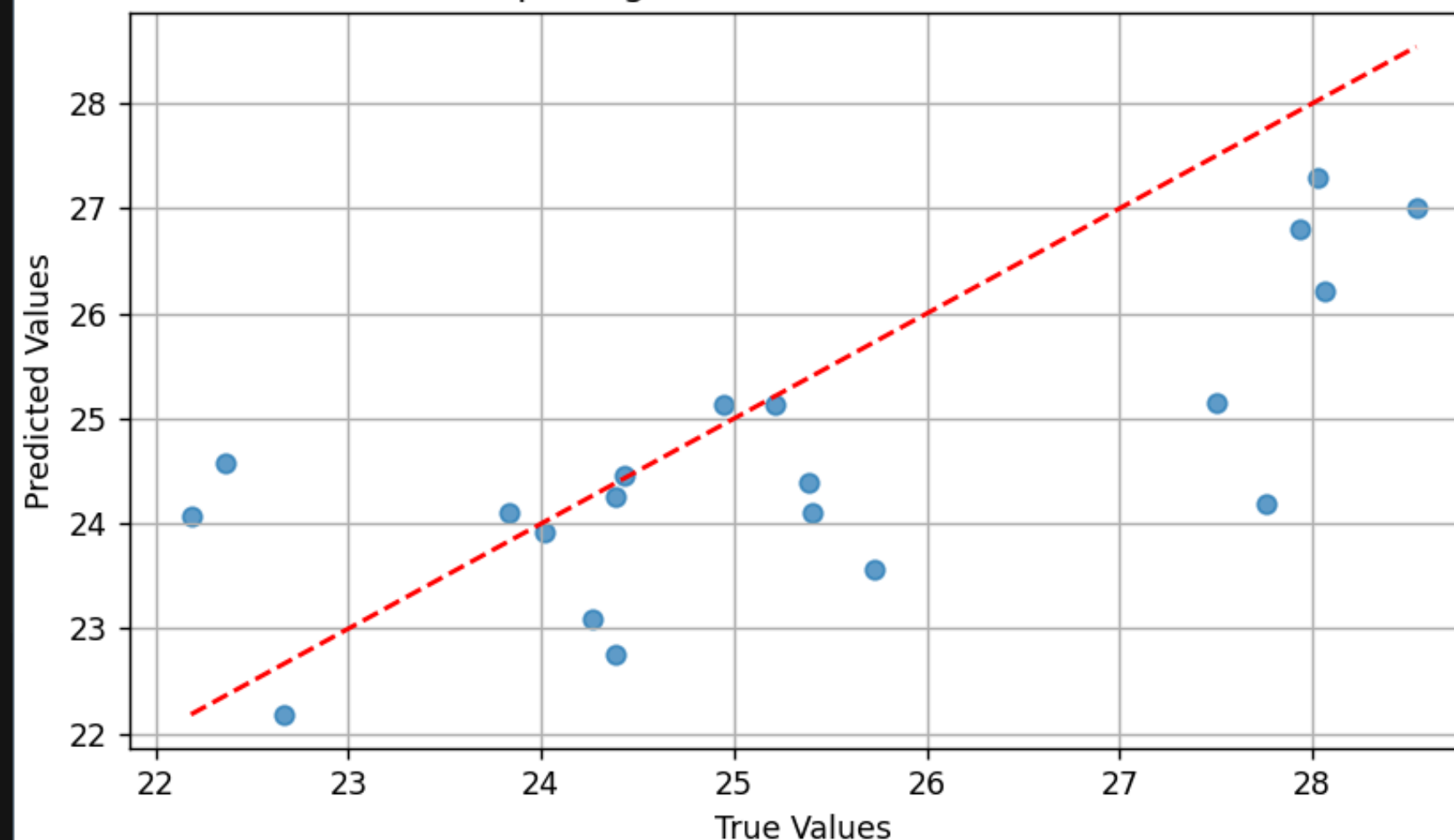
Graph 18 → Predicted: 24.0629 | True: 22.1834 | Abs Error: 1.8795

Graph 2 → Predicted: 23.9181 | True: 24.0147 | Abs Error: 0.0966

Graph 6 → Predicted: 25.1342 | True: 24.9460 | Abs Error: 0.1882

Figure 1

Graph Regression: Predicted vs Actual



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- graph_regression_gra...
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- graph_classification.py
- graph_generation.py
- graph_matching.py
- graph_reconstruction.py
- graph_regression.py

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rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)

\$ bash codeB/run.sh gcn synthetic_regression graph_regression
Running task: graph_regression using dataset: synthetic_regression

No saved model found. Training from scratch...

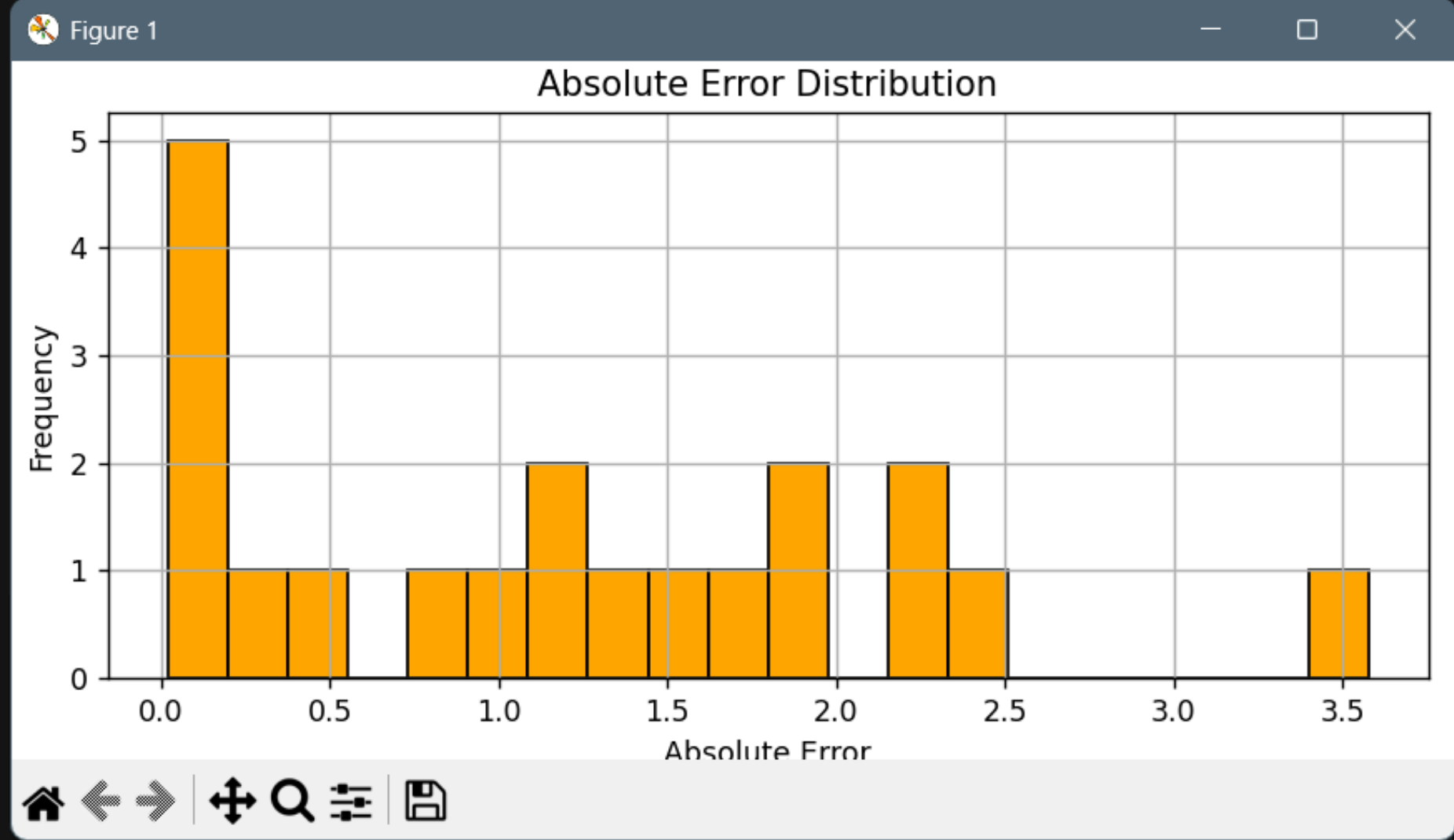
[GCN-GraphReg] Epoch 0 - Train MSE: 97.183635
[GCN-GraphReg] Epoch 10 - Train MSE: 5.703851
[GCN-GraphReg] Epoch 20 - Train MSE: 4.431348
[GCN-GraphReg] Epoch 30 - Train MSE: 5.535823
[GCN-GraphReg] Epoch 40 - Train MSE: 3.090857
[GCN-GraphReg] Epoch 50 - Train MSE: 3.097077
[GCN-GraphReg] Epoch 60 - Train MSE: 2.237914
[GCN-GraphReg] Epoch 70 - Train MSE: 1.971153
[GCN-GraphReg] Epoch 80 - Train MSE: 1.883828
[GCN-GraphReg] Epoch 90 - Train MSE: 2.248514
[GCN-GraphReg] Epoch 99 - Train MSE: 1.946905

Evaluation on Test Set:

MSE: 2.3075
MAE: 1.1903
R² Score: 0.3953

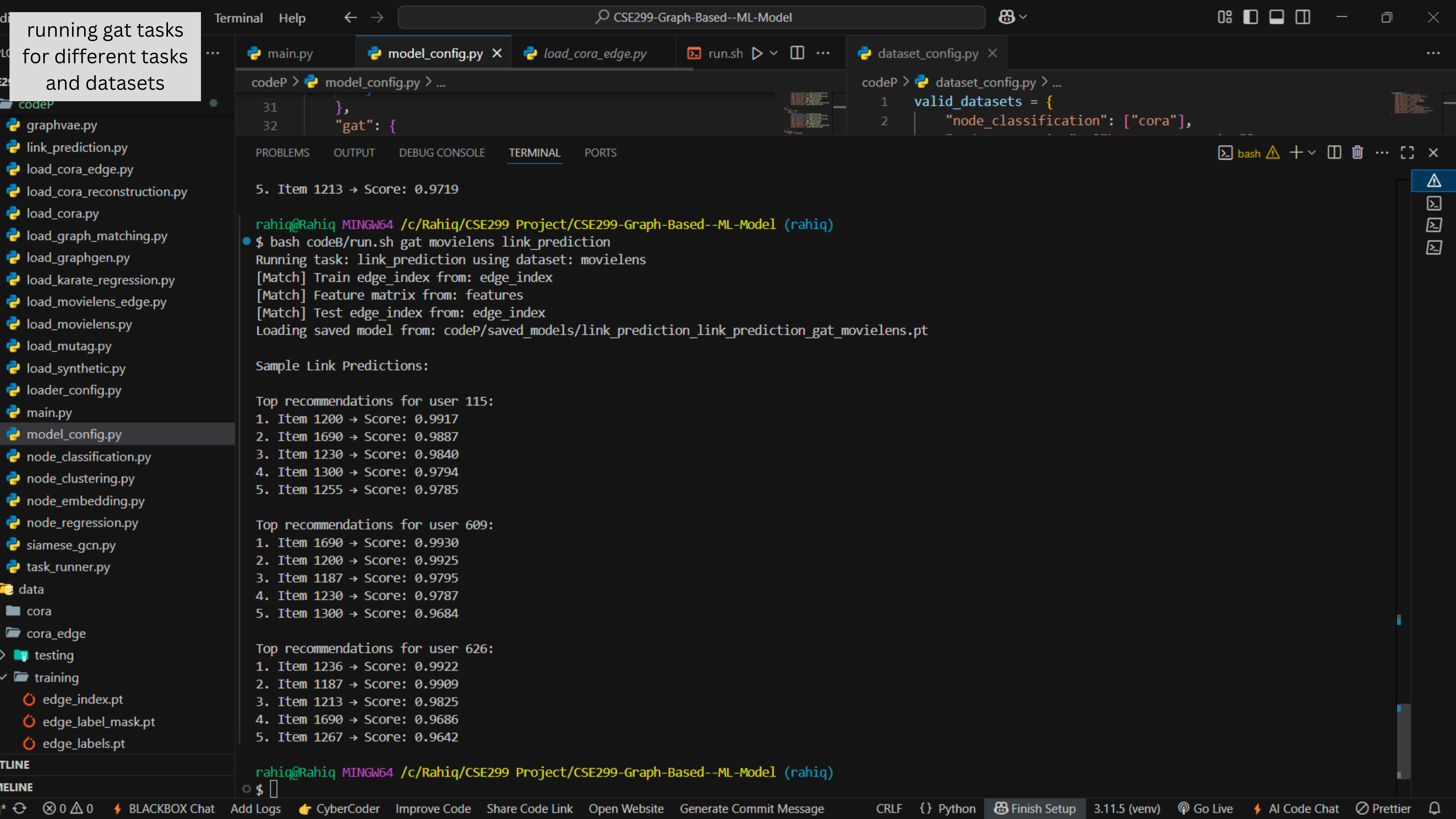
Sample Graph Regression:

Graph 17 → Predicted: 24.0982 True: 23.8365 Abs Error: 0.2618
Graph 13 → Predicted: 25.1579 True: 27.5053 Abs Error: 2.3475
Graph 18 → Predicted: 24.0629 True: 22.1834 Abs Error: 1.8795
Graph 2 → Predicted: 23.9181 True: 24.0147 Abs Error: 0.0966
Graph 6 → Predicted: 25.1342 True: 24.9460 Abs Error: 0.1882



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```
codeP > model_config.py > ...  
31 },  
32 "gat": {
```

```
codeP > dataset_config.py > ...  
1 valid_datasets = {  
2     "node_classification": ["cora"],
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
5. Item 1213 → Score: 0.9719  
  
rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)  
$ bash codeB/run.sh gat movielens link_prediction  
Running task: link_prediction using dataset: movielens  
[Match] Train edge_index from: edge_index  
[Match] Feature matrix from: features  
[Match] Test edge_index from: edge_index  
Loading saved model from: codeP/saved_models/link_prediction_link_prediction_gat_movielens.pt  
  
Sample Link Predictions:  
  
Top recommendations for user 115:  
1. Item 1200 → Score: 0.9917  
2. Item 1690 → Score: 0.9887  
3. Item 1230 → Score: 0.9840  
4. Item 1300 → Score: 0.9794  
5. Item 1255 → Score: 0.9785  
  
Top recommendations for user 609:  
1. Item 1690 → Score: 0.9930  
2. Item 1200 → Score: 0.9925  
3. Item 1187 → Score: 0.9795  
4. Item 1230 → Score: 0.9787  
5. Item 1300 → Score: 0.9684  
  
Top recommendations for user 626:  
1. Item 1236 → Score: 0.9922  
2. Item 1187 → Score: 0.9909  
3. Item 1213 → Score: 0.9825  
4. Item 1690 → Score: 0.9686  
5. Item 1267 → Score: 0.9642
```

```
rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)  
$
```


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- graphvae.py
 - link_prediction.py
 - load_cora_edge.py
 - load_cora_reconstruction.py
 - load_cora.py
 - load_graph_matching.py
 - load_graphgen.py
 - load_karate_regression.py
 - load_movielsens_edge.py
 - load_movielsens.py
 - load_mutag.py
 - load_synthetic.py
 - loader_config.py
 - main.py
 - model_config.py
 - node_classification.py
 - node_clustering.py
 - node_embedding.py
 - node_regression.py
 - siamese_gcn.py
 - task_runner.py
- data
- cora
 - cora_edge
- testing
- training
- edge_index.pt
 - edge_label_mask.pt
 - edge_labels.pt

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh gat cora node_classification
Running task: node_classification using dataset: cora
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] labels from: labels
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
Loading saved model from: codeP/saved_models/node_classification_node_classification_gat_cora.pt

Evaluation on Test Set:
Accuracy: 0.7480
Classification Report:

```

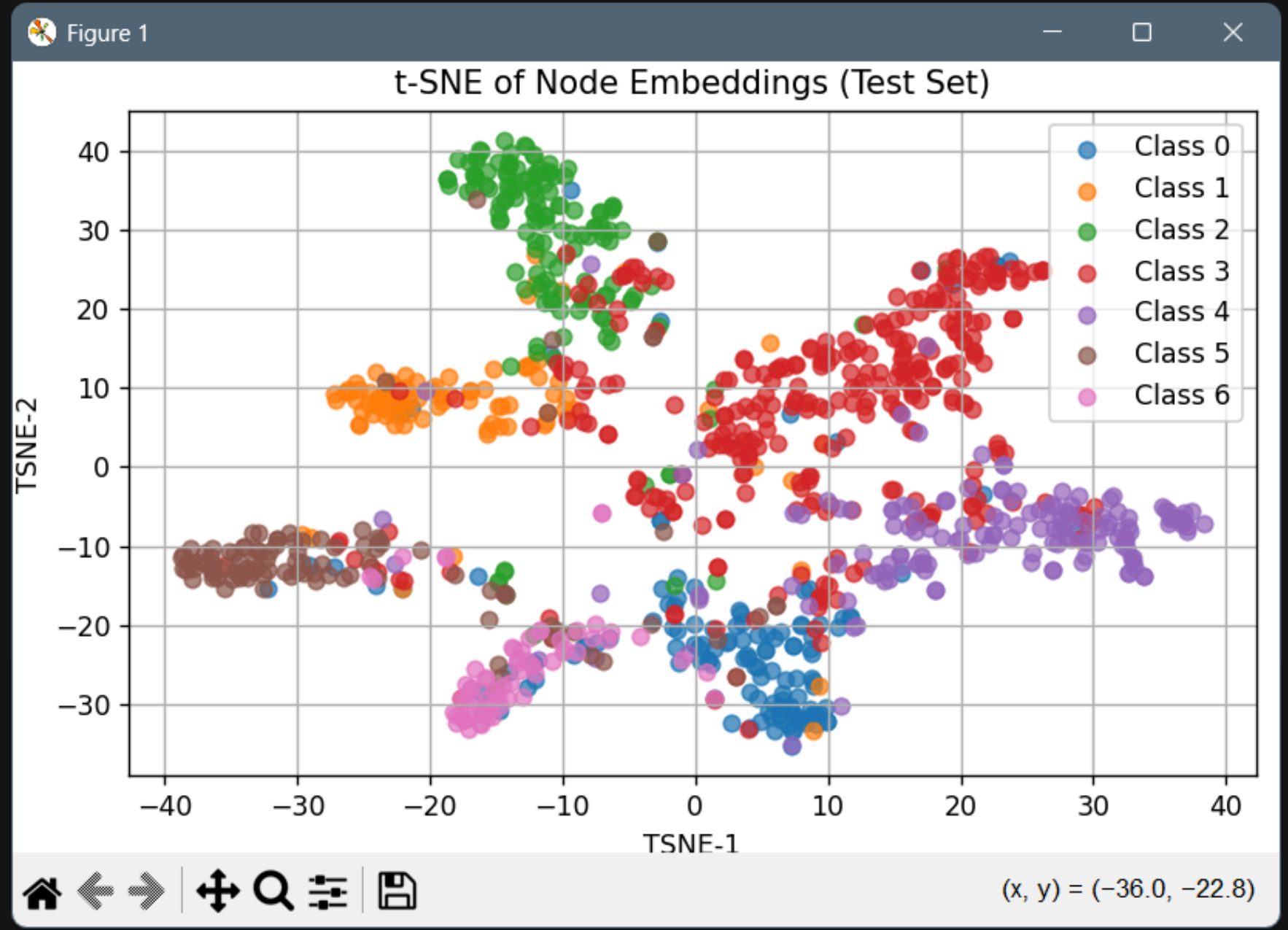
	precision	recall	f1-score	support
0	0.62	0.78	0.69	130
1	0.71	0.86	0.78	91
2	0.83	0.87	0.85	144
3	0.90	0.59	0.71	319
4	0.67	0.88	0.76	149
5	0.81	0.70	0.75	103
6	0.63	0.81	0.71	64
accuracy			0.75	1000
macro avg	0.74	0.78	0.75	1000
weighted avg	0.77	0.75	0.75	1000

Sample Node Predictions (Test Set):

- Node 2549 → Predicted: 4, Actual: 4 ✓
- Node 1970 → Predicted: 4, Actual: 4 ✓
- Node 1891 → Predicted: 4, Actual: 4 ✓
- Node 2609 → Predicted: 1, Actual: 3 ✗
- Node 2150 → Predicted: 3, Actual: 5 ✗

Per-Class Accuracy:

- Class 0: Accuracy = 0.7769
- Class 1: Accuracy = 0.8571
- Class 2: Accuracy = 0.8681
- Class 3: Accuracy = 0.5925
- Class 4: Accuracy = 0.8792



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```
3. Item 1213 → Score: 0.9825
4. Item 1690 → Score: 0.9686
5. Item 1267 → Score: 0.9642
```

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
```

```
$ bash codeB/run.sh gat karate_regression node_regression
```

```
Running task: node_regression using dataset: karate_regression
```

```
[Match] edge_index from: edge_index
```

```
[Match] Feature matrix from: features
```

```
[Match] targets from: targets
```

```
[Match] train_mask from: train_mask
```

```
[Match] test_mask from: test_mask
```

```
No saved model found. Training from scratch...
```

```
[GAT-NodeReg] Epoch 0 - Loss: 14.6798 - Test MSE: 4.6133
```

```
[GAT-NodeReg] Epoch 20 - Loss: 6.2059 - Test MSE: 0.7000
```

```
[GAT-NodeReg] Epoch 40 - Loss: 5.7694 - Test MSE: 1.6028
```

```
[GAT-NodeReg] Epoch 60 - Loss: 5.4134 - Test MSE: 1.4508
```

```
[GAT-NodeReg] Epoch 80 - Loss: 4.9268 - Test MSE: 1.6581
```

```
[GAT-NodeReg] Epoch 100 - Loss: 4.0424 - Test MSE: 1.8954
```

```
[GAT-NodeReg] Epoch 120 - Loss: 1.7378 - Test MSE: 3.1898
```

```
[GAT-NodeReg] Epoch 140 - Loss: 1.0595 - Test MSE: 5.6663
```

```
[GAT-NodeReg] Epoch 160 - Loss: 0.8685 - Test MSE: 6.8621
```

```
[GAT-NodeReg] Epoch 180 - Loss: 0.7795 - Test MSE: 7.0485
```

```
Evaluation on Test Set:
```

```
Mean Squared Error (MSE): 7.7710
```

```
Mean Absolute Error (MAE): 2.3420
```

```
R2 Score: -19.3787
```

```
Sample Node Regression Results (Test Set):
```

```
Node 5 → Predicted: 1.0982 | Actual: 2.3529 | Absolute Error: 1.2548
```

```
Node 30 → Predicted: 4.9602 | Actual: 2.3529 | Absolute Error: 2.6073
```

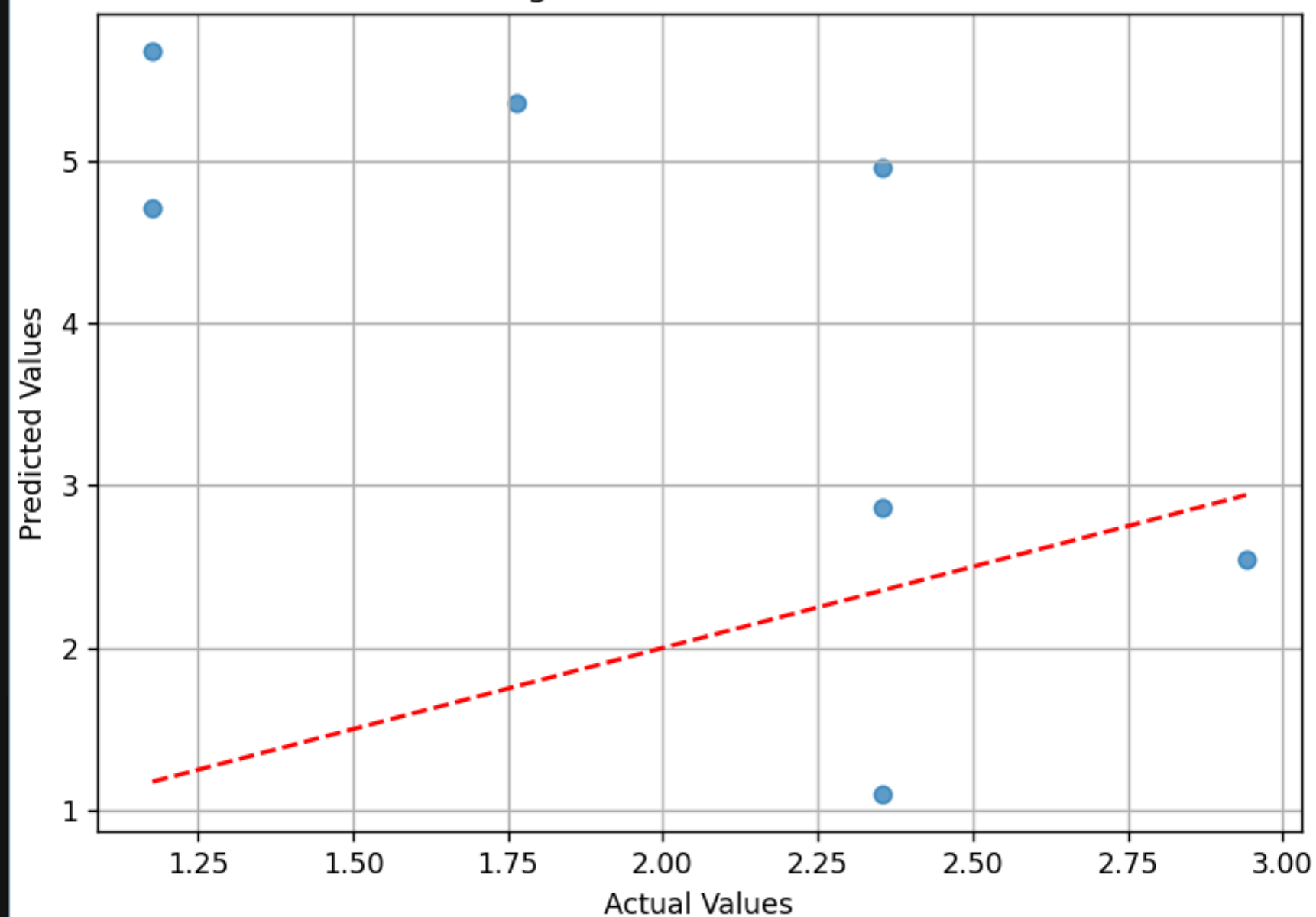
```
Node 13 → Predicted: 2.5404 | Actual: 2.9412 | Absolute Error: 0.4008
```

```
Node 27 → Predicted: 2.8643 | Actual: 2.3529 | Absolute Error: 0.5114
```

```
Node 22 → Predicted: 4.7046 | Actual: 1.1765 | Absolute Error: 3.5281
```

Figure 1

Node Regression: Predicted vs Actual



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```
3. Item 1213 → Score: 0.9825
4. Item 1690 → Score: 0.9686
5. Item 1267 → Score: 0.9642
```

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
```

```
$ bash codeB/run.sh gat karate_regression node_regression
Running task: node_regression using dataset: karate_regression
```

```
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] targets from: targets
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
```

```
No saved model found. Training from scratch...
```

```
[GAT-NodeReg] Epoch 0 - Loss: 14.6798 - Test MSE: 4.6133
```

```
[GAT-NodeReg] Epoch 20 - Loss: 6.2059 - Test MSE: 0.7000
```

```
[GAT-NodeReg] Epoch 40 - Loss: 5.7694 - Test MSE: 1.6028
```

```
[GAT-NodeReg] Epoch 60 - Loss: 5.4134 - Test MSE: 1.4508
```

```
[GAT-NodeReg] Epoch 80 - Loss: 4.9268 - Test MSE: 1.6581
```

```
[GAT-NodeReg] Epoch 100 - Loss: 4.0424 - Test MSE: 1.8954
```

```
[GAT-NodeReg] Epoch 120 - Loss: 1.7378 - Test MSE: 3.1898
```

```
[GAT-NodeReg] Epoch 140 - Loss: 1.0595 - Test MSE: 5.6663
```

```
[GAT-NodeReg] Epoch 160 - Loss: 0.8685 - Test MSE: 6.8621
```

```
[GAT-NodeReg] Epoch 180 - Loss: 0.7795 - Test MSE: 7.0485
```

```
Evaluation on Test Set:
```

```
Mean Squared Error (MSE): 7.7710
```

```
Mean Absolute Error (MAE): 2.3420
```

```
R2 Score: -19.3787
```

```
Sample Node Regression Results (Test Set):
```

```
Node 5 → Predicted: 1.0982 | Actual: 2.3529 | Absolute Error: 1.2548
```

```
Node 30 → Predicted: 4.9602 | Actual: 2.3529 | Absolute Error: 2.6073
```

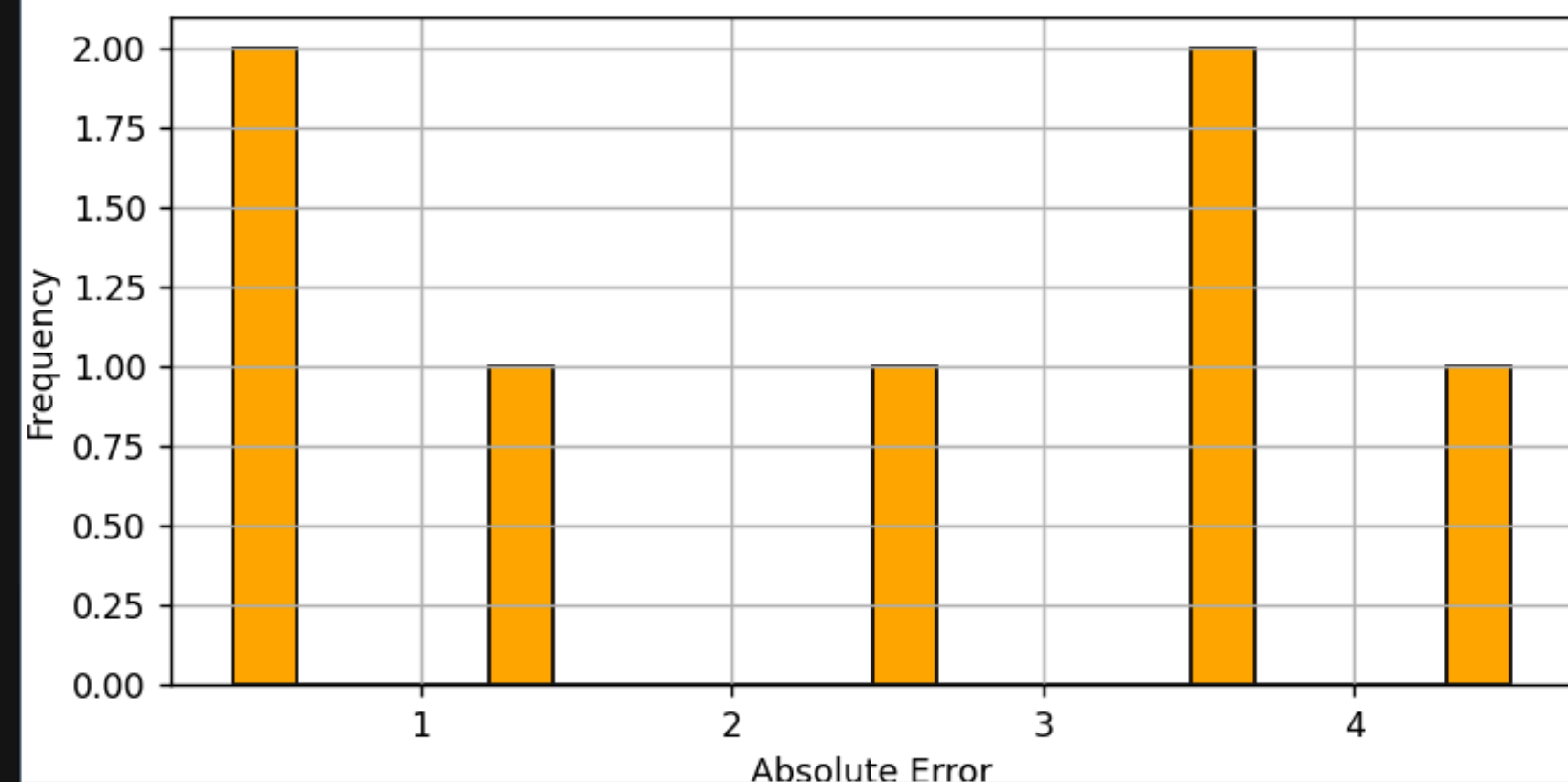
```
Node 13 → Predicted: 2.5404 | Actual: 2.9412 | Absolute Error: 0.4008
```

```
Node 27 → Predicted: 2.8643 | Actual: 2.3529 | Absolute Error: 0.5114
```

```
Node 22 → Predicted: 4.7046 | Actual: 1.1765 | Absolute Error: 3.5281
```

Figure 1

Distribution of Absolute Errors



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(x, y) = (0.896, 0.618)

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weighted avg 0.94 0.94 0.94 2112

Sample Predictions on Test Edges:
Edge (1136, 2359) → Predicted: 1, True: 1, Confidence: 0.9987 ✓
Edge (2205, 48) → Predicted: 1, True: 1, Confidence: 1.0000 ✓
Edge (1339, 1358) → Predicted: 1, True: 1, Confidence: 1.0000 ✓
Edge (459, 1623) → Predicted: 1, True: 1, Confidence: 1.0000 ✓
Edge (1434, 445) → Predicted: 1, True: 1, Confidence: 1.0000 ✓

rahik@Rahik MINGW64 /c/Rahik/CSE299 Project/CSE299-Graph-Based--ML-Model (rahik)
\$ bash codeB/run.sh gat cora_edge edge_classification
Running task: edge_classification using dataset: cora_edge
[Match] edge_index from: edge_index
[Match] edge_labels from: edge_labels
[Match] train_mask from: edge_label_mask
[Match] Feature matrix from: features
[Match] test_mask from: edge_label_mask

Test Accuracy: 0.9432

Classification Report:

	precision	recall	f1-score	support
0	0.84	0.86	0.85	397
1	0.97	0.96	0.96	1715
accuracy			0.94	2112
macro avg	0.90	0.91	0.91	2112
weighted avg	0.94	0.94	0.94	2112

Sample Predictions on Test Edges:
Edge (645, 1900) → Predicted: 0, True: 0, Confidence: 1.0000 ✓
Edge (218, 1020) → Predicted: 1, True: 1, Confidence: 1.0000 ✓
Edge (1896, 1897) → Predicted: 1, True: 1, Confidence: 1.0000 ✓
Edge (884, 152) → Predicted: 1, True: 1, Confidence: 1.0000 ✓
Edge (1171, 687) → Predicted: 0, True: 0, Confidence: 0.7828 ✓

rahik@Rahik MINGW64 /c/Rahik/CSE299 Project/CSE299-Graph-Based--ML-Model (rahik)
\$

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Visualizing node embeddings with t-SNE (Test Set)...

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh gat cora node_clustering

Running task: node_clustering using dataset: cora

No saved model found. Training from scratch...

[GAT-NodeEmbed] Epoch 0 - MSE Loss: 0.0137

[GAT-NodeEmbed] Epoch 20 - MSE Loss: 0.0106

[GAT-NodeEmbed] Epoch 40 - MSE Loss: 0.0095

[GAT-NodeEmbed] Epoch 60 - MSE Loss: 0.0088

[GAT-NodeEmbed] Epoch 80 - MSE Loss: 0.0083

Total Clusters: 7

Cluster Size Distribution:

Cluster 2: 520 nodes

Cluster 1: 557 nodes

Cluster 4: 920 nodes

Cluster 6: 312 nodes

Cluster 5: 393 nodes

Cluster 0: 1 nodes

Cluster 3: 5 nodes

Sample Node Clusters:

Node 2082 → Cluster 4

Node 1032 → Cluster 4

Node 750 → Cluster 4

Node 1482 → Cluster 4

Node 233 → Cluster 4

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$

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edge_regression_edge_regr...

graph_classification_graph...

graph_generation_graph_genera...

graph_regression_graph_re...

link_prediction_link_predicti...

link_prediction_link_predicti...

node_classification_node_cl...

node_classification_node_cl...

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node_clustering_node_clust...

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graph_generation_graph_genera...

graph_regression_graph_re...

link_prediction_link_predicti...

link_prediction_link_predicti...

node_classification_node_cl...

node_classification_node_cl...

node_clustering_node_clust...

node_clustering_node_clust...

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node_embedding_node_em...

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Node 233 → Cluster 4

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh gat cora node_embedding

Running task: node_embedding using dataset: cora

No saved model found. Training from scratch...

[GAT-NodeEmbed] Epoch 0 - MSE Loss: 0.0137

[GAT-NodeEmbed] Epoch 20 - MSE Loss: 0.0106

[GAT-NodeEmbed] Epoch 40 - MSE Loss: 0.0096

[GAT-NodeEmbed] Epoch 60 - MSE Loss: 0.0089

[GAT-NodeEmbed] Epoch 80 - MSE Loss: 0.0084

Sample Node Embeddings:

Node 0 → Embedding: [0.0067, 0.0330, 0.0790, -0.0038, -0.0481, ...]

Node 1 → Embedding: [-0.0192, -0.0088, 0.0948, 0.0563, 0.1226, ...]

Node 2 → Embedding: [0.0041, -0.0007, 0.0441, 0.0277, 0.0589, ...]

Node 3 → Embedding: [-0.0036, 0.0033, -0.1214, 0.0114, -0.0461, ...]

Node 4 → Embedding: [-0.0083, 0.1102, -0.0547, 0.0874, 0.0038, ...]

Embedding Statistics:

Mean: 0.0129

Std Dev: 0.0673

Min: -0.3890

Max: 1.5086

Cosine Similarity Between Sample Nodes:

Node 0 vs Node 1: 0.3207

Node 0 vs Node 2: 0.3361

Node 0 vs Node 3: 0.0785

Node 0 vs Node 4: 0.2491

Node 1 vs Node 2: 0.7050

Node 1 vs Node 3: 0.1858

Node 1 vs Node 4: 0.2944

Node 2 vs Node 3: 0.2607

Node 2 vs Node 4: 0.2876

Node 3 vs Node 4: 0.0639

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```
Node 0 vs Node 4: 0.2491
Node 1 vs Node 2: 0.7050
Node 1 vs Node 3: 0.1858
Node 1 vs Node 4: 0.2944
Node 2 vs Node 3: 0.2607
Node 2 vs Node 4: 0.2876
Node 3 vs Node 4: 0.0639
```

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
```

```
$ bash codeB/run.sh gat synthetic_regression graph_regression
Running task: graph_regression using dataset: synthetic_regression
```

```
No saved model found. Training from scratch...
```

```
[GAT-GraphReg] Epoch 0 - Train MSE: 43.769165
```

```
[GAT-GraphReg] Epoch 10 - Train MSE: 4.442048
```

```
[GAT-GraphReg] Epoch 20 - Train MSE: 2.307533
```

```
[GAT-GraphReg] Epoch 30 - Train MSE: 2.729299
```

```
[GAT-GraphReg] Epoch 40 - Train MSE: 1.827737
```

```
[GAT-GraphReg] Epoch 50 - Train MSE: 2.056783
```

```
[GAT-GraphReg] Epoch 60 - Train MSE: 1.245393
```

```
[GAT-GraphReg] Epoch 70 - Train MSE: 1.990289
```

```
[GAT-GraphReg] Epoch 80 - Train MSE: 1.284598
```

```
[GAT-GraphReg] Epoch 90 - Train MSE: 0.932240
```

```
[GAT-GraphReg] Epoch 99 - Train MSE: 1.143649
```

```
Evaluation on Test Set:
```

```
MSE: 2.1997
```

```
MAE: 1.2271
```

```
R2 Score: 0.4236
```

```
Sample Graph Regression:
```

```
Graph 3 → Predicted: 24.6267 | True: 24.3823 | Abs Error: 0.2444
```

```
Graph 19 → Predicted: 23.6177 | True: 22.3638 | Abs Error: 1.2539
```

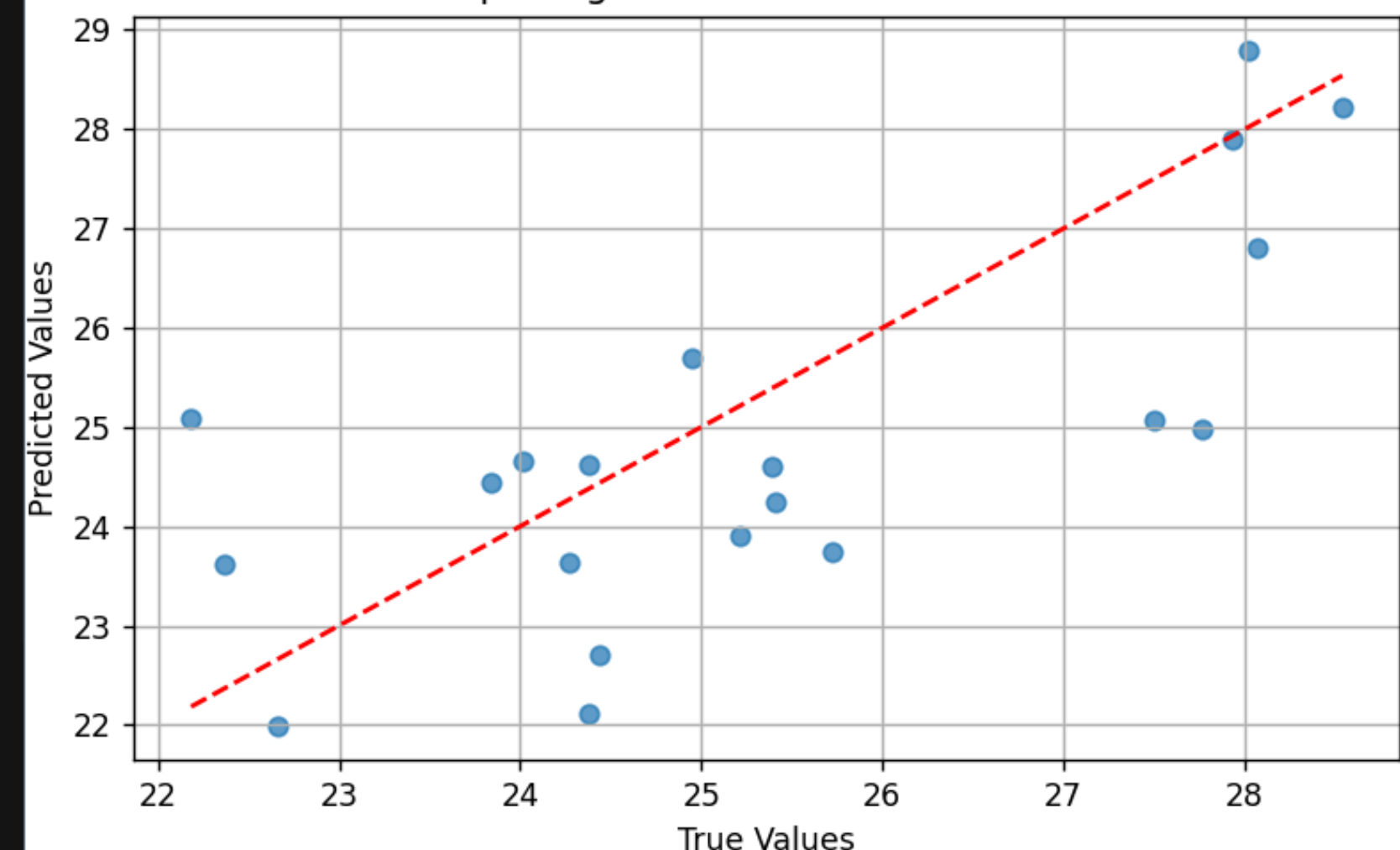
```
Graph 16 → Predicted: 22.7005 | True: 24.4355 | Abs Error: 1.7350
```

```
Graph 18 → Predicted: 25.0921 | True: 22.1834 | Abs Error: 2.9087
```

```
Graph 10 → Predicted: 23.9009 | True: 25.2133 | Abs Error: 1.3124
```

Figure 1

Graph Regression: Predicted vs Actual



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Node 0 vs Node 4: 0.2491
Node 1 vs Node 2: 0.7050
Node 1 vs Node 3: 0.1858
Node 1 vs Node 4: 0.2944
Node 2 vs Node 3: 0.2607
Node 2 vs Node 4: 0.2876
Node 3 vs Node 4: 0.0639
```

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
```

```
$ bash codeB/run.sh gat synthetic_regression graph_regression
```

```
Running task: graph_regression using dataset: synthetic_regression
```

```
No saved model found. Training from scratch...
```

```
[GAT-GraphReg] Epoch 0 - Train MSE: 43.769165
```

```
[GAT-GraphReg] Epoch 10 - Train MSE: 4.442048
```

```
[GAT-GraphReg] Epoch 20 - Train MSE: 2.307533
```

```
[GAT-GraphReg] Epoch 30 - Train MSE: 2.729299
```

```
[GAT-GraphReg] Epoch 40 - Train MSE: 1.827737
```

```
[GAT-GraphReg] Epoch 50 - Train MSE: 2.056783
```

```
[GAT-GraphReg] Epoch 60 - Train MSE: 1.245393
```

```
[GAT-GraphReg] Epoch 70 - Train MSE: 1.990289
```

```
[GAT-GraphReg] Epoch 80 - Train MSE: 1.284598
```

```
[GAT-GraphReg] Epoch 90 - Train MSE: 0.932240
```

```
[GAT-GraphReg] Epoch 99 - Train MSE: 1.143649
```

```
Evaluation on Test Set:
```

```
MSE: 2.1997
```

```
MAE: 1.2271
```

```
R2 Score: 0.4236
```

```
Sample Graph Regression:
```

```
Graph 3 → Predicted: 24.6267 | True: 24.3823 | Abs Error: 0.2444
```

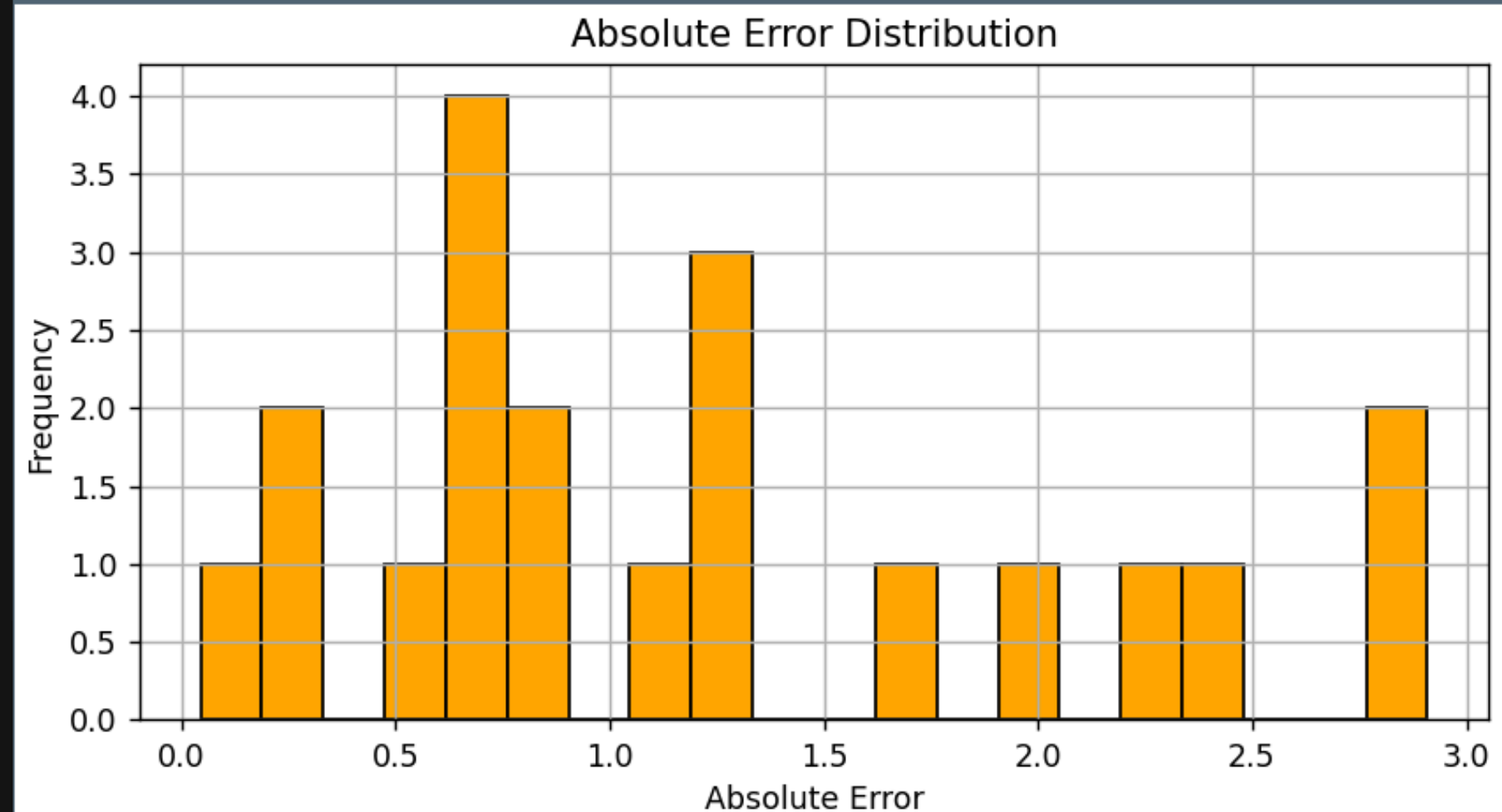
```
Graph 19 → Predicted: 23.6177 | True: 22.3638 | Abs Error: 1.2539
```

```
Graph 16 → Predicted: 22.7005 | True: 24.4355 | Abs Error: 1.7350
```

```
Graph 18 → Predicted: 25.0921 | True: 22.1834 | Abs Error: 2.9087
```

```
Graph 10 → Predicted: 23.9009 | True: 25.2133 | Abs Error: 1.3124
```

Figure 1



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	1	0.75	0.60	0.67	25
accuracy				0.61	38
macro avg		0.60	0.61	0.59	38
weighted avg		0.65	0.61	0.62	38

Sample Graph Classification:

Graph 26 → Predicted: 1 | Actual: 1 ✓

Graph 14 → Predicted: 1 | Actual: 1 ✓

Graph 3 → Predicted: 1 | Actual: 1 ✓

Graph 21 → Predicted: 1 | Actual: 0 ✗

Graph 0 → Predicted: 0 | Actual: 1 ✗

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh gat mutag graph_classification

Running task: graph_classification using dataset: mutag

Loading saved model from: codeP/saved_models/graph_classification_graph_classification_gat_mutag.pt

Evaluation on Test Set:

Accuracy: 0.6053

Classification Report:

	precision	recall	f1-score	support
0	0.44	0.62	0.52	13
1	0.75	0.60	0.67	25
accuracy			0.61	38
macro avg	0.60	0.61	0.59	38
weighted avg	0.65	0.61	0.62	38

Sample Graph Classification:

Graph 12 → Predicted: 0 | Actual: 0 ✓

Graph 31 → Predicted: 0 | Actual: 1 ✗

Graph 26 → Predicted: 1 | Actual: 1 ✓

Graph 3 → Predicted: 1 | Actual: 1 ✓

Graph 13 → Predicted: 1 | Actual: 0 ✗

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

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rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)

\$ bash codeB/run.sh graphsage movielens link_prediction

Running task: link_prediction using dataset: movielens

[Match] Train edge_index from: edge_index

[Match] Feature matrix from: features

[Match] Test edge_index from: edge_index

Loading saved model from: codeP/saved_models/link_prediction_link_prediction_graphsage_movielens.pt

Sample Link Predictions:

Top recommendations for user 806:

1. Item 1230 → Score: 0.8887

2. Item 1042 → Score: 0.8877

3. Item 1200 → Score: 0.8852

4. Item 992 → Score: 0.8756

5. Item 1236 → Score: 0.8737

Top recommendations for user 287:

1. Item 1200 → Score: 0.9322

2. Item 1236 → Score: 0.9166

3. Item 1228 → Score: 0.9126

4. Item 1230 → Score: 0.9044

5. Item 1242 → Score: 0.8660

Top recommendations for user 716:

1. Item 1228 → Score: 0.9106

2. Item 1242 → Score: 0.9074

3. Item 1230 → Score: 0.8902

4. Item 1200 → Score: 0.8777

5. Item 1255 → Score: 0.8612

rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)

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```
$ bash codeB/run.sh graphsage cora node_classification
Running task: node_classification using dataset: cora
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] labels from: labels
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
Loading saved model from: codeP/saved_models/node_classification_node_classification_graphsage_cora.pt
```

Evaluation on Test Set:

Accuracy: 0.7510

Classification Report:

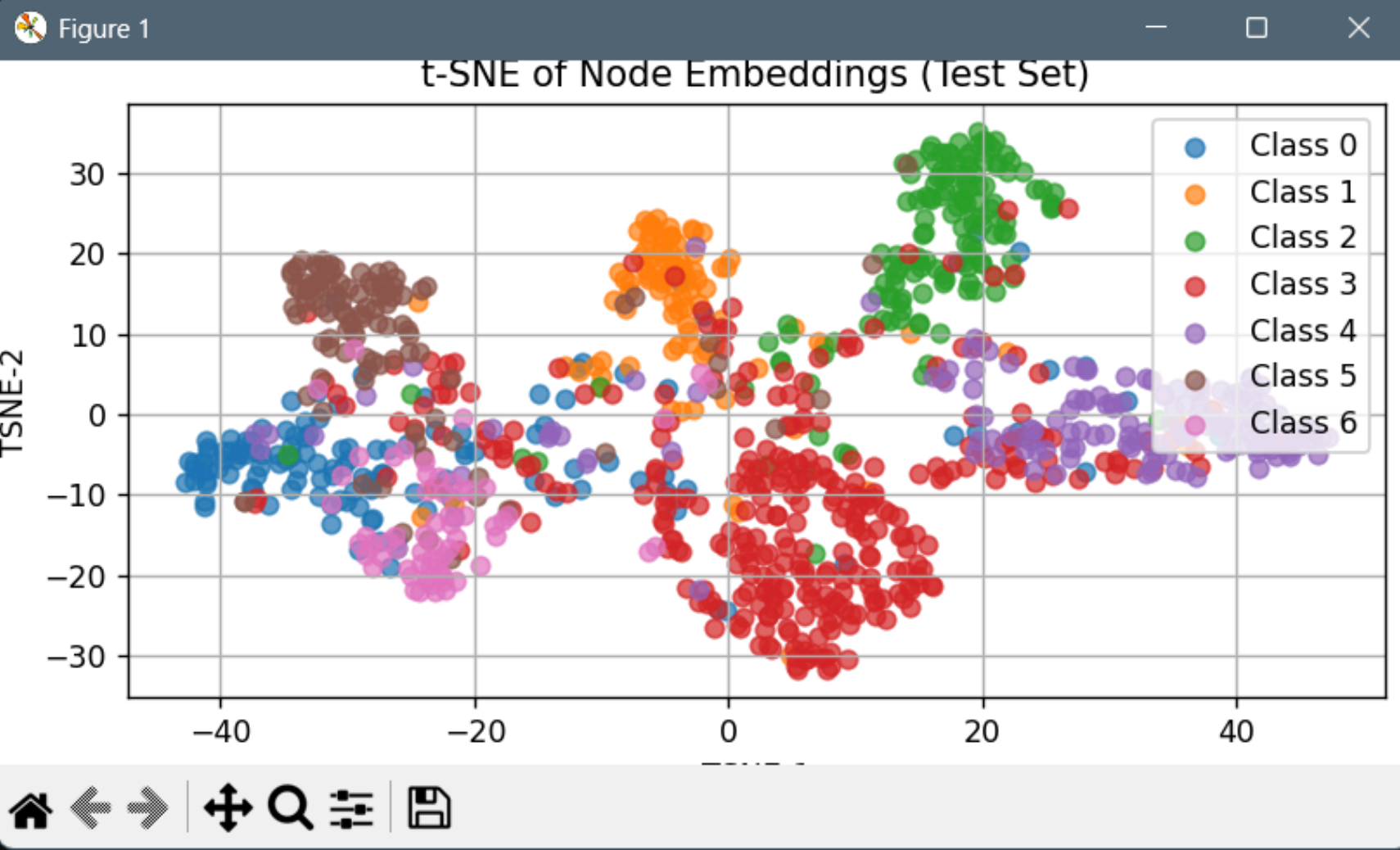
	precision	recall	f1-score	support
0	0.71	0.61	0.65	130
1	0.66	0.85	0.74	91
2	0.76	0.89	0.82	144
3	0.90	0.66	0.76	319
4	0.75	0.87	0.81	149
5	0.80	0.74	0.77	103
6	0.50	0.81	0.62	64
accuracy			0.75	1000
macro avg	0.73	0.77	0.74	1000
weighted avg	0.78	0.75	0.75	1000

Sample Node Predictions (Test Set):

```
Node 2207 → Predicted: 3, Actual: 3 ✓
Node 1830 → Predicted: 2, Actual: 2 ✓
Node 1756 → Predicted: 2, Actual: 2 ✓
Node 1933 → Predicted: 3, Actual: 3 ✓
Node 2027 → Predicted: 0, Actual: 3 ✗
```

Per-Class Accuracy:

```
Class 0: Accuracy = 0.6077
Class 1: Accuracy = 0.8462
Class 2: Accuracy = 0.8889
Class 3: Accuracy = 0.6552
Class 4: Accuracy = 0.8725
```



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"graphsage": {

1

valid_datasets = {

Edge (1976, 502) → Predicted: 1, True: 0, Confidence: 0.9996

Edge (487, 1773) → Predicted: 1, True: 1, Confidence: 1.0000

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh graphsage cora_edge edge_classification

Running task: edge_classification using dataset: cora_edge

[Match] edge_index from: edge_index

[Match] edge_labels from: edge_labels

[Match] train_mask from: edge_label_mask

[Match] Feature matrix from: features

[Match] test_mask from: edge_label_mask

Test Accuracy: 0.9238

Classification Report:

	precision	recall	f1-score	support
0	0.78	0.83	0.80	397
1	0.96	0.95	0.95	1715
accuracy			0.92	2112
macro avg	0.87	0.89	0.88	2112
weighted avg	0.93	0.92	0.92	2112

Sample Predictions on Test Edges:

Edge (108, 1647) → Predicted: 1, True: 1, Confidence: 1.0000

Edge (306, 294) → Predicted: 1, True: 1, Confidence: 0.9778

Edge (306, 236) → Predicted: 1, True: 1, Confidence: 0.9740

Edge (1331, 1810) → Predicted: 1, True: 1, Confidence: 1.0000

Edge (2639, 917) → Predicted: 1, True: 1, Confidence: 1.0000

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

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\$ bash codeB/run.sh graphsage karate_regression

Task '' not supported by model 'graphsage'

rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)

\$ bash codeB/run.sh graphsage karate_regression node_regression

Running task: node_regression using dataset: karate_regression

[Match] edge_index from: edge_index

[Match] Feature matrix from: features

[Match] targets from: targets

[Match] train_mask from: train_mask

[Match] test_mask from: test_mask

No saved model found. Training from scratch...

[GraphSAGE-NodeReg] Epoch 0 - Loss: 15.6977 - Test MSE: 5.4115

[GraphSAGE-NodeReg] Epoch 20 - Loss: 12.6047 - Test MSE: 3.3598

[GraphSAGE-NodeReg] Epoch 40 - Loss: 7.2330 - Test MSE: 0.4917

[GraphSAGE-NodeReg] Epoch 60 - Loss: 5.2988 - Test MSE: 1.7305

[GraphSAGE-NodeReg] Epoch 80 - Loss: 3.9239 - Test MSE: 1.3625

[GraphSAGE-NodeReg] Epoch 100 - Loss: 2.3611 - Test MSE: 1.4057

[GraphSAGE-NodeReg] Epoch 120 - Loss: 1.0268 - Test MSE: 1.5307

[GraphSAGE-NodeReg] Epoch 140 - Loss: 0.4340 - Test MSE: 2.2240

[GraphSAGE-NodeReg] Epoch 160 - Loss: 0.2113 - Test MSE: 2.8163

[GraphSAGE-NodeReg] Epoch 180 - Loss: 0.0982 - Test MSE: 3.1200

Evaluation on Test Set:

Mean Squared Error (MSE): 3.2534

Mean Absolute Error (MAE): 1.1627

R² Score: -7.5318

Sample Node Regression Results (Test Set):

Node 24 → Predicted: 6.0732 | Actual: 1.7647 | Absolute Error: 4.3085

Node 13 → Predicted: 3.0006 | Actual: 2.9412 | Absolute Error: 0.0595

Node 22 → Predicted: 1.8364 | Actual: 1.1765 | Absolute Error: 0.6599

Node 30 → Predicted: 3.8766 | Actual: 2.3529 | Absolute Error: 1.5237

Node 5 → Predicted: 2.7137 | Actual: 2.3529 | Absolute Error: 0.3607

Figure 1

Node Regression: Predicted vs Actual

Actual Values	Predicted Values
1.1	2.3
1.1	1.8
1.8	6.1
2.3	3.9
2.3	2.7
2.3	2.3
2.9	3.0

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run.sh
dataset_config.py

codeP > model_config.py > ...
codeP > dataset_config.py > ...
1 valid datasets = {

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rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)
\$ bash codeB/run.sh graphsage karate_regression
Task '' not supported by model 'graphsage'

rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)
\$ bash codeB/run.sh graphsage karate_regression node_regression
Running task: node_regression using dataset: karate_regression
[Match] edge_index from: edge_index
[Match] Feature matrix from: features
[Match] targets from: targets
[Match] train_mask from: train_mask
[Match] test_mask from: test_mask
No saved model found. Training from scratch...
[GraphSAGE-NodeReg] Epoch 0 - Loss: 15.6977 - Test MSE: 5.4115
[GraphSAGE-NodeReg] Epoch 20 - Loss: 12.6047 - Test MSE: 3.3598
[GraphSAGE-NodeReg] Epoch 40 - Loss: 7.2330 - Test MSE: 0.4917
[GraphSAGE-NodeReg] Epoch 60 - Loss: 5.2988 - Test MSE: 1.7305
[GraphSAGE-NodeReg] Epoch 80 - Loss: 3.9239 - Test MSE: 1.3625
[GraphSAGE-NodeReg] Epoch 100 - Loss: 2.3611 - Test MSE: 1.4057
[GraphSAGE-NodeReg] Epoch 120 - Loss: 1.0268 - Test MSE: 1.5307
[GraphSAGE-NodeReg] Epoch 140 - Loss: 0.4340 - Test MSE: 2.2240
[GraphSAGE-NodeReg] Epoch 160 - Loss: 0.2113 - Test MSE: 2.8163
[GraphSAGE-NodeReg] Epoch 180 - Loss: 0.0982 - Test MSE: 3.1200

Evaluation on Test Set:
Mean Squared Error (MSE): 3.2534
Mean Absolute Error (MAE): 1.1627
R² Score: -7.5318

Sample Node Regression Results (Test Set):
Node 24 → Predicted: 6.0732 | Actual: 1.7647 | Absolute Error: 4.
Node 13 → Predicted: 3.0006 | Actual: 2.9412 | Absolute Error: 0.
Node 22 → Predicted: 1.8364 | Actual: 1.1765 | Absolute Error: 0.
Node 30 → Predicted: 3.8766 | Actual: 2.3529 | Absolute Error: 1.5237
Node 5 → Predicted: 2.7137 | Actual: 2.3529 | Absolute Error: 0.3607
█

Figure 1
Distribution of Absolute Errors

Absolute Error Range	Frequency
0.0 - 0.25	2
0.25 - 0.5	1
0.5 - 0.75	1
1.0 - 1.25	1
1.25 - 1.5	1
4.0 - 4.25	1

0 1 2 3 4

Frequency

Absolute Error

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rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)

\$ bash codeB/run.sh graphsage cora node_clustering

Running task: node_clustering using dataset: cora

No saved model found. Training from scratch...

[SAGE-NodeEmbed] Epoch 0 - MSE Loss: 0.0197

[SAGE-NodeEmbed] Epoch 20 - MSE Loss: 0.0122

[SAGE-NodeEmbed] Epoch 40 - MSE Loss: 0.0120

[SAGE-NodeEmbed] Epoch 60 - MSE Loss: 0.0119

[SAGE-NodeEmbed] Epoch 80 - MSE Loss: 0.0119

Total Clusters: 7

Cluster Size Distribution:

Cluster 3: 347 nodes

Cluster 6: 404 nodes

Cluster 2: 508 nodes

Cluster 4: 567 nodes

Cluster 5: 188 nodes

Cluster 0: 241 nodes

Cluster 1: 453 nodes

Sample Node Clusters:

Node 2510 → Cluster 6

Node 1899 → Cluster 1

Node 1925 → Cluster 1

Node 810 → Cluster 2

Node 1373 → Cluster 5

rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)

\$

graph_reconstruction.py

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graphsage.py

graphvae.py

link_prediction.py

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load_cora_reconstruction.py

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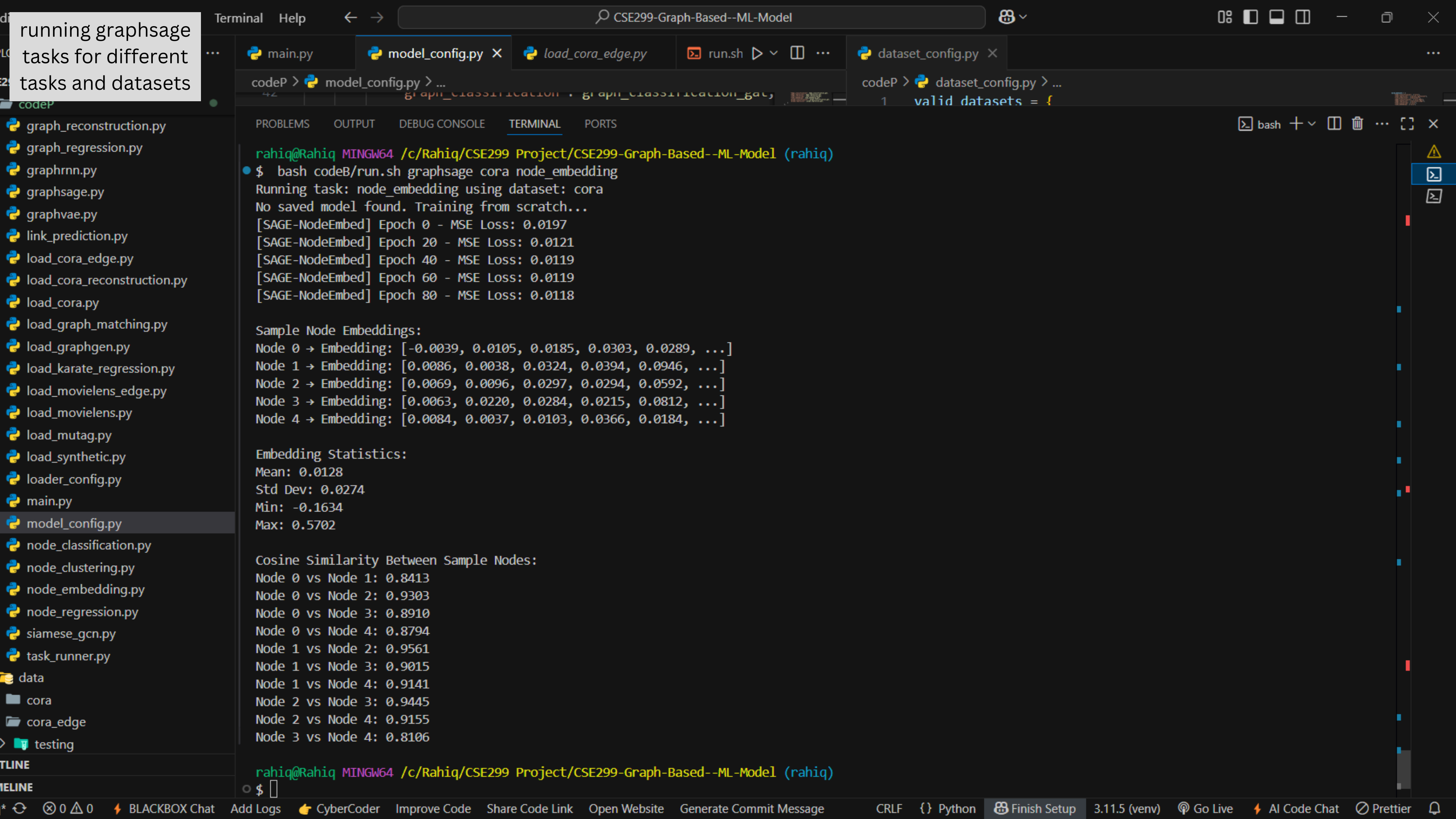
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\$ bash codeB/run.sh graphsage mutag graph_classification

Running task: graph_classification using dataset: mutag

No saved model found. Training from scratch...

[SAGE-GraphCls] Epoch 0 - Loss: 0.6692 - Train Acc: 0.6267

[SAGE-GraphCls] Epoch 10 - Loss: 0.4789 - Train Acc: 0.7800

[SAGE-GraphCls] Epoch 20 - Loss: 0.4687 - Train Acc: 0.7600

[SAGE-GraphCls] Epoch 30 - Loss: 0.4500 - Train Acc: 0.7933

[SAGE-GraphCls] Epoch 40 - Loss: 0.4559 - Train Acc: 0.7733

[SAGE-GraphCls] Epoch 50 - Loss: 0.4612 - Train Acc: 0.7867

[SAGE-GraphCls] Epoch 60 - Loss: 0.4556 - Train Acc: 0.7733

[SAGE-GraphCls] Epoch 70 - Loss: 0.4566 - Train Acc: 0.7933

[SAGE-GraphCls] Epoch 80 - Loss: 0.4539 - Train Acc: 0.7867

[SAGE-GraphCls] Epoch 90 - Loss: 0.4498 - Train Acc: 0.7667

[SAGE-GraphCls] Epoch 99 - Loss: 0.4411 - Train Acc: 0.8067

Evaluation on Test Set:

Accuracy: 0.7368

Classification Report:

	precision	recall	f1-score	support
0	0.71	0.38	0.50	13
1	0.74	0.92	0.82	25
accuracy			0.74	38
macro avg	0.73	0.65	0.66	38
weighted avg	0.73	0.74	0.71	38

Sample Graph Classification:

Graph 8 → Predicted: 0 | Actual: 0 ✓

Graph 20 → Predicted: 1 | Actual: 1 ✓

Graph 23 → Predicted: 1 | Actual: 1 ✓

Graph 11 → Predicted: 1 | Actual: 1 ✓

Graph 2 → Predicted: 1 | Actual: 1 ✓

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Node 2510 → Cluster 6

Node 1899 → Cluster 1

Node 1925 → Cluster 1

Node 810 → Cluster 2

Node 1373 → Cluster 5

rahig@Rahig MINGW64 /c/Rahig/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh graphsage synthetic_regression graph_regression

Running task: graph_regression using dataset: synthetic_regression

No saved model found. Training from scratch...

[SAGE-GraphReg] Epoch 0 - Train MSE: 59.439233

[SAGE-GraphReg] Epoch 10 - Train MSE: 0.879258

[SAGE-GraphReg] Epoch 20 - Train MSE: 0.308367

[SAGE-GraphReg] Epoch 30 - Train MSE: 0.060316

[SAGE-GraphReg] Epoch 40 - Train MSE: 0.146401

[SAGE-GraphReg] Epoch 50 - Train MSE: 0.153610

[SAGE-GraphReg] Epoch 60 - Train MSE: 0.059445

[SAGE-GraphReg] Epoch 70 - Train MSE: 0.074574

[SAGE-GraphReg] Epoch 80 - Train MSE: 0.059787

[SAGE-GraphReg] Epoch 90 - Train MSE: 0.103218

[SAGE-GraphReg] Epoch 99 - Train MSE: 0.019562

Evaluation on Test Set:

MSE: 0.0033

MAE: 0.0466

R² Score: 0.9991

Sample Graph Regression:

Graph 0 → Predicted: 24.2384 | True: 24.2668 | Abs Error: 0.0285

Graph 10 → Predicted: 25.1781 | True: 25.2133 | Abs Error: 0.0352

Graph 8 → Predicted: 24.3987 | True: 24.3831 | Abs Error: 0.0156

Graph 18 → Predicted: 22.1987 | True: 22.1834 | Abs Error: 0.0153

Graph 9 → Predicted: 25.7524 | True: 25.7283 | Abs Error: 0.0241

Figure 1

Graph Regression: Predicted vs Actual

True Values	Predicted Values
22.1834	22.1987
22.2668	24.2384
24.3831	24.3987
24.2133	25.1781
25.7283	25.7524
25.2133	25.1781

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Node 2510 → Cluster 6

Node 1899 → Cluster 1

Node 1925 → Cluster 1

Node 810 → Cluster 2

Node 1373 → Cluster 5

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh graphsage synthetic_regression graph_regr

Running task: graph_regression using dataset: synthetic_regre

No saved model found. Training from scratch...

[SAGE-GraphReg] Epoch 0 - Train MSE: 59.439233

[SAGE-GraphReg] Epoch 10 - Train MSE: 0.879258

[SAGE-GraphReg] Epoch 20 - Train MSE: 0.308367

[SAGE-GraphReg] Epoch 30 - Train MSE: 0.060316

[SAGE-GraphReg] Epoch 40 - Train MSE: 0.146401

[SAGE-GraphReg] Epoch 50 - Train MSE: 0.153610

[SAGE-GraphReg] Epoch 60 - Train MSE: 0.059445

[SAGE-GraphReg] Epoch 70 - Train MSE: 0.074574

[SAGE-GraphReg] Epoch 80 - Train MSE: 0.059787

[SAGE-GraphReg] Epoch 90 - Train MSE: 0.103218

[SAGE-GraphReg] Epoch 99 - Train MSE: 0.019562

Evaluation on Test Set:

MSE: 0.0033

MAE: 0.0466

R² Score: 0.9991

Sample Graph Regression:

Graph 0 → Predicted: 24.2384 | True: 24.2668 | Abs Error: 0.0

Graph 10 → Predicted: 25.1781 | True: 25.2133 | Abs Error: 0.

Graph 8 → Predicted: 24.3987 | True: 24.3831 | Abs Error: 0.0

Graph 18 → Predicted: 22.1987 | True: 22.1834 | Abs Error: 0.

Graph 9 → Predicted: 25.7524 | True: 25.7283 | Abs Error: 0.0241

Figure 1

Absolute Error Distribution

Absolute Error Range	Frequency
0.01 - 0.02	2.0
0.02 - 0.03	3.0
0.03 - 0.04	3.0
0.04 - 0.05	2.0
0.05 - 0.06	2.0
0.06 - 0.07	2.0
0.07 - 0.08	1.0
0.08 - 0.09	3.0
0.09 - 0.10	1.0
0.10 - 0.11	0.0
0.11 - 0.12	0.0
0.12 - 0.13	0.0
0.13 - 0.14	1.0
0.14 - 0.15	1.0

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Task 'graph_generation' not supported by model 'gae'

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based-ML-Model (working)

\$ bash codeB/run.sh gae cora_reconstruction graph_reconstruction

Running task: graph_reconstruction using dataset: cora_reconstruction

[Match] Feature matrix from: x

[Match] Train edge_index from: train_edge_index

[Match] Test edge_index from: test_edge_index

Graph Statistics:

Number of nodes: 2708

Number of edges: 9500

Average degree: 7.02

Max degree: 167.0

No saved model found. Training from scratch...

Epoch 10/100 - Loss: 1.2820

Epoch 20/100 - Loss: 1.0184

Epoch 30/100 - Loss: 0.9255

Epoch 40/100 - Loss: 0.8849

Epoch 50/100 - Loss: 0.8509

Epoch 60/100 - Loss: 0.8338

Epoch 70/100 - Loss: 0.8288

Epoch 80/100 - Loss: 0.8133

Epoch 90/100 - Loss: 0.7903

Epoch 100/100 - Loss: 0.8025

Graph Reconstruction Results:

AUC: 0.9558

Average Precision (AP): 0.9542

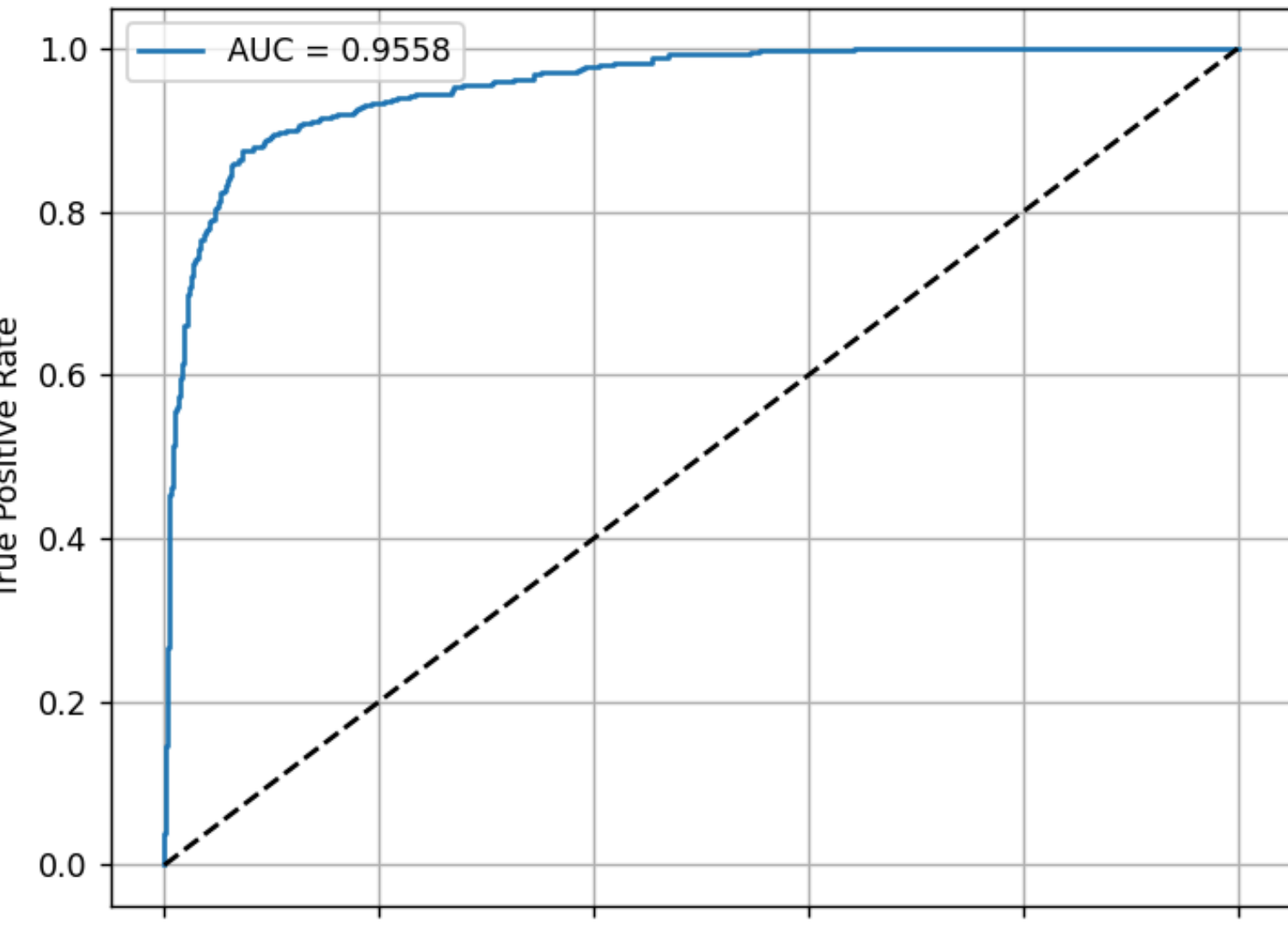
F1-Score (threshold 0.5): 0.8249

Reconstruction Accuracy (threshold 0.5): 0.7926

Figure 1

ROC Curve

AUC = 0.9558



True Positive Rate

False Positive Rate

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Task 'graph_generation' not supported by model 'gae'

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Gr
\$ bash codeB/run.sh gae cora_reconstruction graph_reco
Running task: graph_reconstruction using dataset: cora
[Match] Feature matrix from: x
[Match] Train edge_index from: train_edge_index
[Match] Test edge_index from: test_edge_index

Graph Statistics:
Number of nodes: 2708
Number of edges: 9500
Average degree: 7.02
Max degree: 167.0
No saved model found. Training from scratch...
Epoch 10/100 - Loss: 1.2820
Epoch 20/100 - Loss: 1.0184
Epoch 30/100 - Loss: 0.9255
Epoch 40/100 - Loss: 0.8849
Epoch 50/100 - Loss: 0.8509
Epoch 60/100 - Loss: 0.8338
Epoch 70/100 - Loss: 0.8288
Epoch 80/100 - Loss: 0.8133
Epoch 90/100 - Loss: 0.7903
Epoch 100/100 - Loss: 0.8025

Graph Reconstruction Results:
AUC: 0.9558
Average Precision (AP): 0.9542
F1-Score (threshold 0.5): 0.8249
Reconstruction Accuracy (threshold 0.5): 0.7926
█

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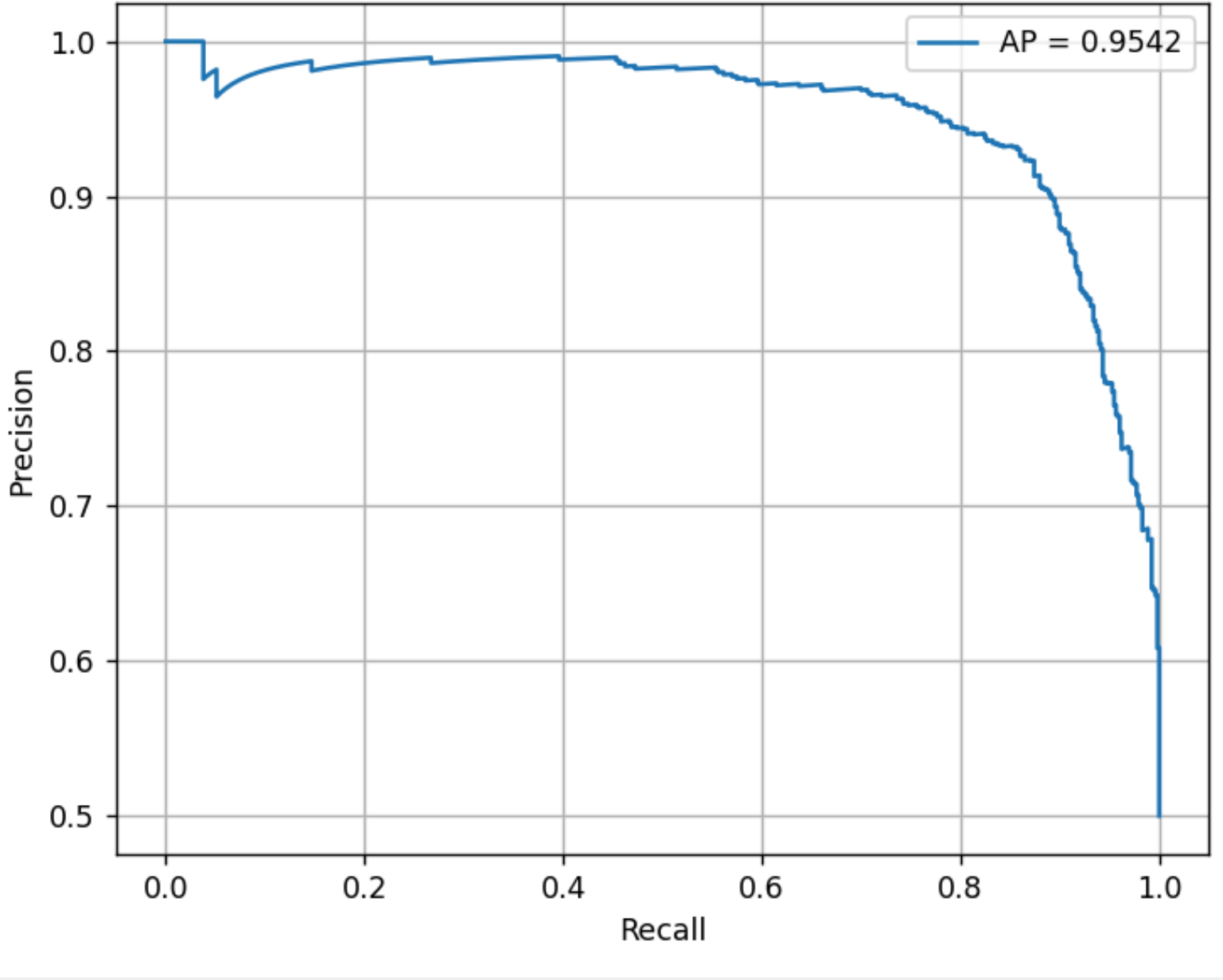
graph_generation.py

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Figure 1

Precision-Recall Curve



AP = 0.9542

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Task 'graph_generation' not supported by model 'gae'

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based

\$ bash codeB/run.sh gae cora_reconstruction graph_reconstruction

Running task: graph_reconstruction using dataset: cora_reconstruction

[Match] Feature matrix from: x

[Match] Train edge_index from: train_edge_index

[Match] Test edge_index from: test_edge_index

Graph Statistics:

Number of nodes: 2708

Number of edges: 9500

Average degree: 7.02

Max degree: 167.0

No saved model found. Training from scratch...

Epoch 10/100 - Loss: 1.2820

Epoch 20/100 - Loss: 1.0184

Epoch 30/100 - Loss: 0.9255

Epoch 40/100 - Loss: 0.8849

Epoch 50/100 - Loss: 0.8509

Epoch 60/100 - Loss: 0.8338

Epoch 70/100 - Loss: 0.8288

Epoch 80/100 - Loss: 0.8133

Epoch 90/100 - Loss: 0.7903

Epoch 100/100 - Loss: 0.8025

Graph Reconstruction Results:

AUC: 0.9558

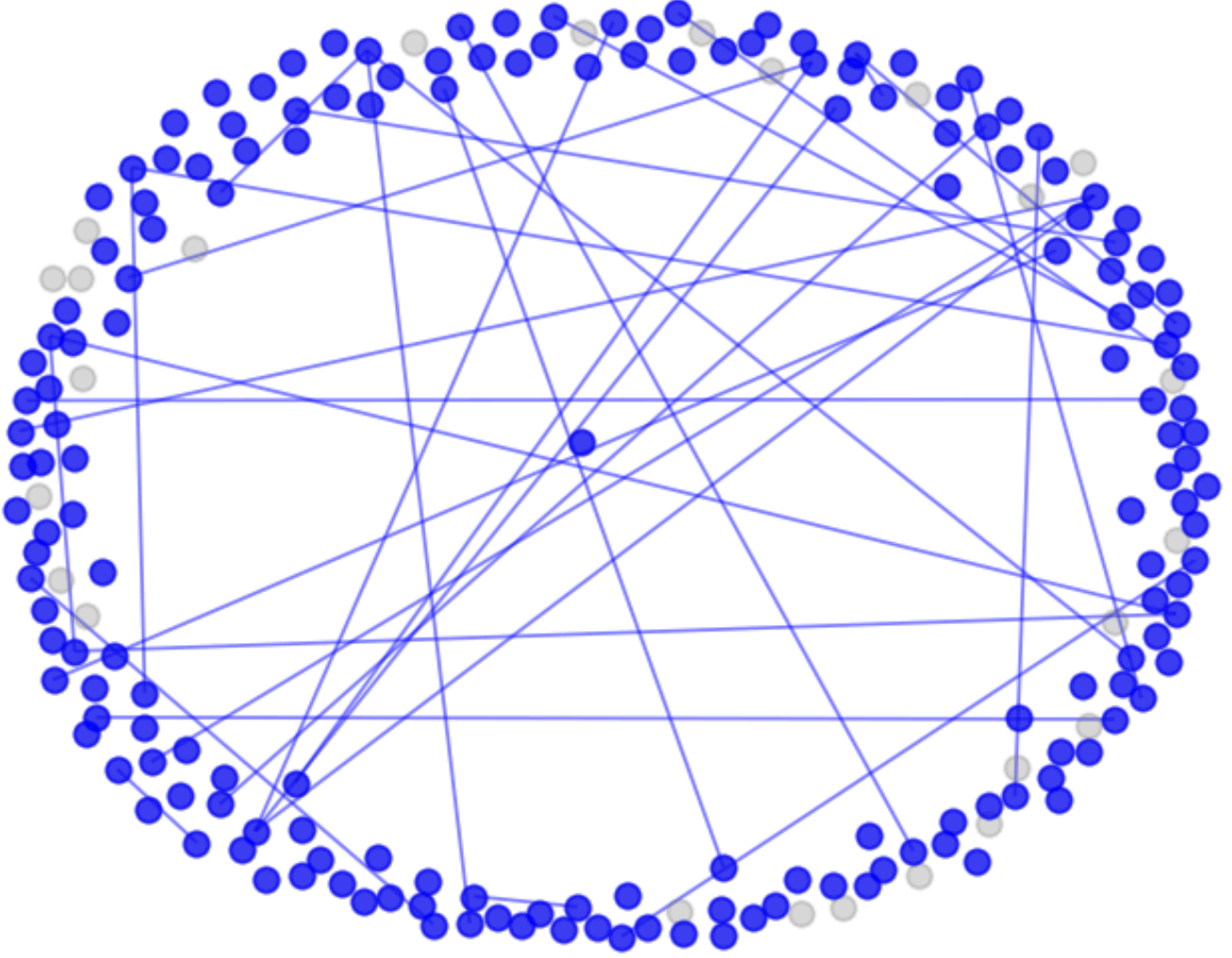
Average Precision (AP): 0.9542

F1-Score (threshold 0.5): 0.8249

Reconstruction Accuracy (threshold 0.5): 0.7926

Figure 1

Original Graph Subgraph



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Task 'graph_generation' not supported by model 'gae'

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph

\$ bash codeB/run.sh gae cora_reconstruction graph_recons

Running task: graph_reconstruction using dataset: cora_r

[Match] Feature matrix from: x

[Match] Train edge_index from: train_edge_index

[Match] Test edge_index from: test_edge_index

Graph Statistics:

Number of nodes: 2708

Number of edges: 9500

Average degree: 7.02

Max degree: 167.0

No saved model found. Training from scratch...

Epoch 10/100 - Loss: 1.2820

Epoch 20/100 - Loss: 1.0184

Epoch 30/100 - Loss: 0.9255

Epoch 40/100 - Loss: 0.8849

Epoch 50/100 - Loss: 0.8509

Epoch 60/100 - Loss: 0.8338

Epoch 70/100 - Loss: 0.8288

Epoch 80/100 - Loss: 0.8133

Epoch 90/100 - Loss: 0.7903

Epoch 100/100 - Loss: 0.8025

Graph Reconstruction Results:

AUC: 0.9558

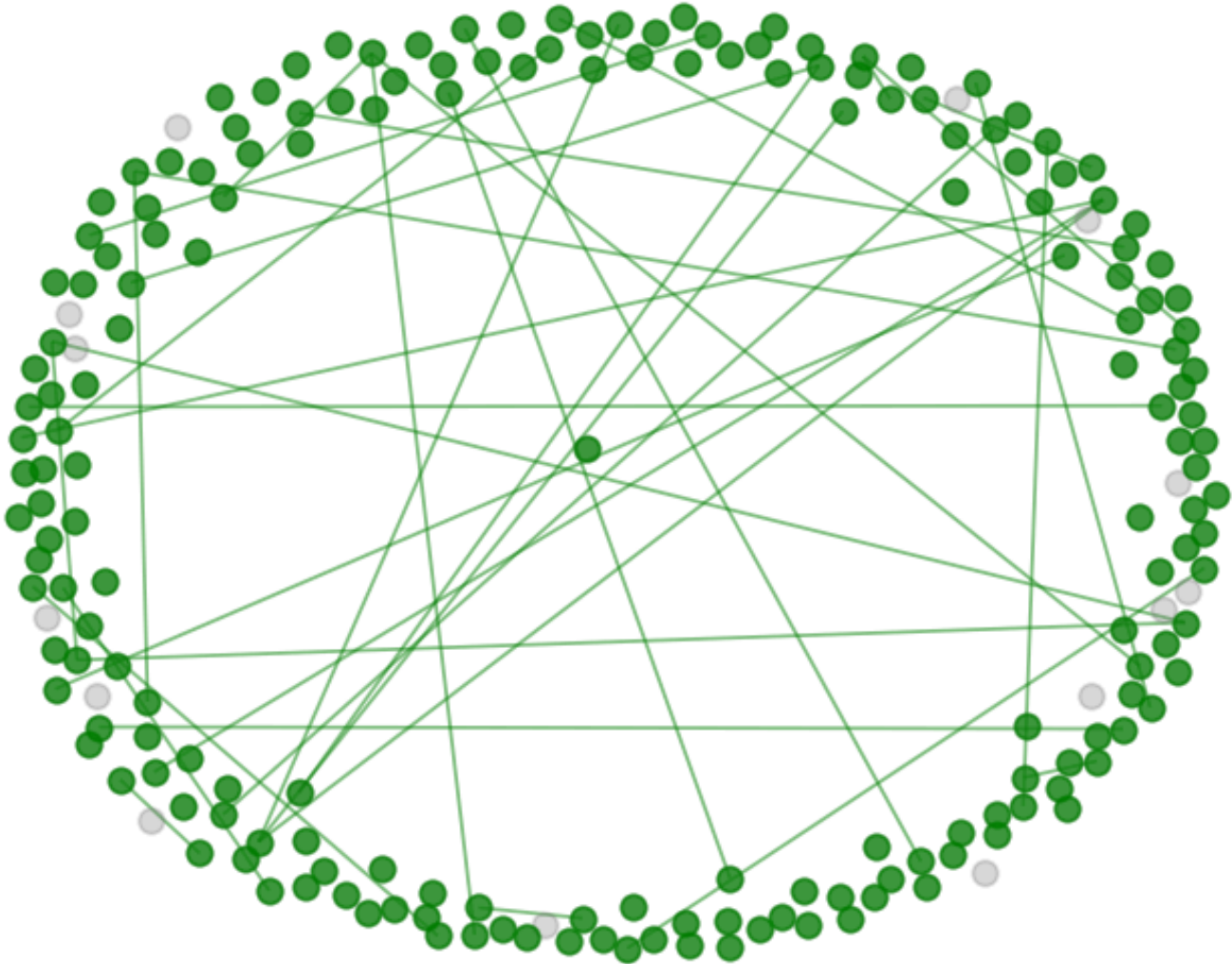
Average Precision (AP): 0.9542

F1-Score (threshold 0.5): 0.8249

Reconstruction Accuracy (threshold 0.5): 0.7926

Figure 1

Reconstructed Graph Subgraph





(x, y) = (0.399, 0.959)

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node_classification_node_cl... U

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grae.py

graph_classification.py

graph_generation.py

Terminal Help

CSE299-Graph-Based--ML-Model

main.py

model_config.py

load_cora_edge.py

run.sh

dataset_config.py

codeP > model_config.py > ...

61 "name": "Siamese GCN",

62 "module": {

63 "graph_matching": graph_matching siamese gcn.

codeP > dataset_config.py > ...

1 valid_datasets = {

2 "node_classification": ["cora"],

3 "node_regression": ["karate_regression"],

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

bash

Pair 67 → Predicted: True | Actual: 1 ✓

Pair 125 → Predicted: True | Actual: 1 ✓

rahig@Rahig MINGW64 /c/Rahig/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh siamese_gcn graph_matching graph_matching

Running task: graph_matching using dataset: graph_matching

Loading saved model from: codeP/saved_models/graph_matching_graph_matching_siamese_gcn_graph_matching.pt

Evaluation on Test Set:

Accuracy: 0.6750

Classification Report:

	precision	recall	f1-score	support
0	0.68	0.47	0.55	86
1	0.67	0.83	0.75	114
accuracy			0.68	200
macro avg	0.68	0.65	0.65	200
weighted avg	0.68	0.68	0.66	200

Sample Graph Matching Results:

Pair 75 → Predicted: False | Actual: 1 ✗

Pair 133 → Predicted: True | Actual: 1 ✓

Pair 24 → Predicted: False | Actual: 0 ✓

Pair 106 → Predicted: True | Actual: 0 ✗

Pair 66 → Predicted: True | Actual: 0 ✗

rahig@Rahig MINGW64 /c/Rahig/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$

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running graphvae tasks for different tasks and datasets

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load_cora_edge.py

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dataset_config.py

codeP > model_config.py > ...

codeP > dataset_config.py > ...

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Pair 133 → Predicted: True | Actual: 1 ✓

Pair 24 → Predicted: False | Actual: 0 ✓

Pair 133 → Predicted: True | Actual: 1 ✓

Pair 133 → Predicted: True | Actual: 1 ✓

Pair 24 → Predicted: False | Actual: 0 ✓

Pair 106 → Predicted: True | Actual: 0 ✗

Pair 66 → Predicted: True | Actual: 0 ✗

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh graphvae graph_generation graph_generation

Running task: graph_generation using dataset: graph_generation

No saved model found. Training from scratch...

Epoch 1/100 - Loss: 217.4662

Epoch 11/100 - Loss: 198.7323

Epoch 21/100 - Loss: 197.8625

Epoch 31/100 - Loss: 197.9491

Epoch 41/100 - Loss: 198.5119

Epoch 51/100 - Loss: 199.2338

Epoch 61/100 - Loss: 197.6290

Epoch 71/100 - Loss: 199.0022

Epoch 81/100 - Loss: 198.3082

Epoch 91/100 - Loss: 198.5532

Epoch 100/100 - Loss: 197.5289

Generated Graphs:

Graph 1 → Nodes: 10 | Edges: 32

Avg Degree: 6.40

Max Degree: 8

task_runner_files

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link_prediction_link_predicti...

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node_classification_node_cl...

node_classification_node_cl...

node_clustering_node_clust...

node_clustering_node_clust...

node_clustering_node_clust...

node_embedding_node_em...

node_embedding_node_em...

node_embedding_node_em...

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node_regression_node_regr...

node_regression_node_regr...

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Figure 1

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dataset_config.py

codeP > model_config.py > ...

codeP > dataset_config.py > ...

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Pair 133 → Predicted: True | Actual: 1 ✓

Pair 24 → Predicted: False | Actual: 0 ✓

Pair 133 → Predicted: True | Actual: 1 ✓

Pair 133 → Predicted: True | Actual: 1 ✓

Pair 24 → Predicted: False | Actual: 0 ✓

Pair 106 → Predicted: True | Actual: 0 ✗

Pair 66 → Predicted: True | Actual: 0 ✗

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh graphvae graph_generation graph_generation

Running task: graph_generation using dataset: graph_generation

No saved model found. Training from scratch...

Epoch 1/100 - Loss: 217.4662

Epoch 11/100 - Loss: 198.7323

Epoch 21/100 - Loss: 197.8625

Epoch 31/100 - Loss: 197.9491

Epoch 41/100 - Loss: 198.5119

Epoch 51/100 - Loss: 199.2338

Epoch 61/100 - Loss: 197.6290

Epoch 71/100 - Loss: 199.0022

Epoch 81/100 - Loss: 198.3082

Epoch 91/100 - Loss: 198.5532

Epoch 100/100 - Loss: 197.5289

Generated Graphs:

Graph 1 → Nodes: 10 | Edges: 32

Avg Degree: 6.40

Max Degree: 8

Graph 2 → Nodes: 10 | Edges: 31

Avg Degree: 6.20

Max Degree: 10

task_runner_files

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edge_regression.py

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gmh.py

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graph_regression_graph_re...

link_prediction_link_predicti...

link_prediction_link_predicti...

link_prediction_link_predicti...

node_classification_node_cl...

node_classification_node_cl...

node_classification_node_cl...

node_clustering_node_clust...

node_clustering_node_clust...

node_clustering_node_clust...

node_embedding_node_em...

node_embedding_node_em...

node_embedding_node_em...

node_regression_node_regr...

node_regression_node_regr...

node_regression_node_regr...

task_runner_files

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gat.py

gcn.py

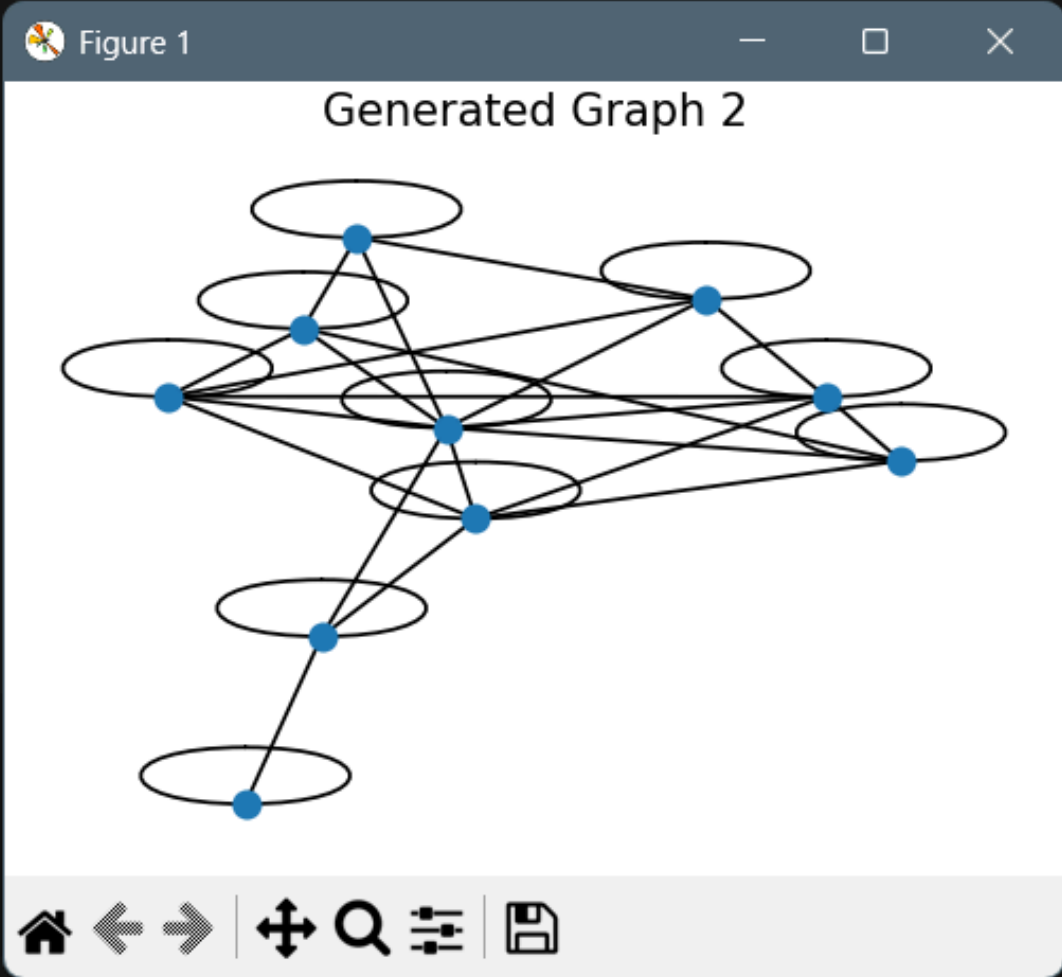
gmh.py

grae.py

graph_classification.py

Figure 1

Generated Graph 2



bash

bash

powershell

bash

bash

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load_cora_edge.py

run.sh

dataset_config.py

codeP > model_config.py > ...

codeP > dataset_config.py > ...

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"grae": {

"module": {

14

}

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PROBLEMS

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Pair 45 → Predicted: True | Actual: 0 ❌

Pair 194 → Predicted: False | Actual: 1 ❌

Pair 45 → Predicted: True | Actual: 0 ❌

Pair 45 → Predicted: True | Actual: 0 ❌

Pair 194 → Predicted: False | Actual: 1 ❌

Pair 135 → Predicted: False | Actual: 0 ✅

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh gmn graph_matching graph_matching

Running task: graph_matching using dataset: graph_matching

Loading saved model from: codeP/saved_models/graph_matching_graph_matching_gmn_graph_matching.pt

Evaluation on Test Set:

Accuracy: 0.6800

Classification Report:

	precision	recall	f1-score	support
0	0.62	0.65	0.64	86
1	0.73	0.70	0.71	114
accuracy			0.68	200
macro avg	0.67	0.68	0.68	200
weighted avg	0.68	0.68	0.68	200

Sample Graph Matching Results:

Pair 107 → Predicted: True | Actual: 1 ✅

Pair 127 → Predicted: True | Actual: 1 ✅

Pair 77 → Predicted: True | Actual: 1 ✅

Pair 9 → Predicted: False | Actual: 0 ✅

Pair 123 → Predicted: True | Actual: 1 ✅

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$

bash

bash

powershell

graph_regression_graph_re...

graph_regression_graph_re...

graph_regression_graph_re...

link_prediction_link_predicti...

link_prediction_link_predicti...

link_prediction_link_predicti...

node_classification_node_cl...

node_classification_node_cl...

node_classification_node_cl...

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node_clustering_node_clust...

node_clustering_node_clust...

node_embedding_node_em...

node_embedding_node_em...

node_embedding_node_em...

node_regression_node_regr...

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codeP > dataset_config.py > ...

83 "graphrnn": {

84 "module": {

PROBLEMS

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Avg Degree: 7.00

Max Degree: 9

rah iq@Rah iq MINGW64 /c/Rah iq/CSE299 Project/CSE299-Graph-Based--ML-Model (rah iq)

\$ bash codeB/run.sh graphrnn graph_generation graph_generation

Running task: graph_generation using dataset: graph_generation

Loading saved model from: codeP/saved_models/graph_generation_graph_generation_graphrnn_graph_generation.pt

Generated Graphs:

Graph 1 → Nodes: 10 | Edges: 45

Avg Degree: 9.00

Max Degree: 9

graph_regression_graph_re...

graph_regression_graph_re...

graph_regression_graph_re...

link_prediction_link_predicti...

link_prediction_link_predicti...

link_prediction_link_predicti...

node_classification_node_cl...

node_classification_node_cl...

node_classification_node_cl...

node_clustering_node_clust...

node_clustering_node_clust...

node_clustering_node_clust...

node_embedding_node_em...

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node_regression_node_regr...

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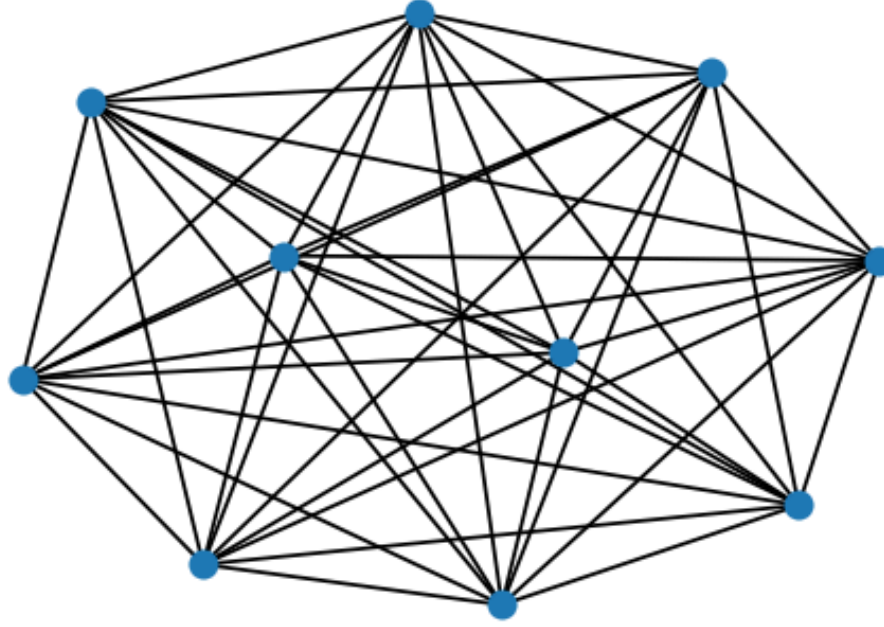
bash

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Figure 1

Generated Graph 1



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graph_regression_graph_re... U

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link_prediction_link_predicti... U

link_prediction_link_predicti... U

node_classification_node_cl... U

node_classification_node_cl... U

node_classification_node_cl... U

node_clustering_node_clust... U

node_clustering_node_clust... U

node_clustering_node_clust... U

node_embedding_node_em... U

node_embedding_node_em... U

node_embedding_node_em... U

node_regression_node_regr... U

node_regression_node_regr... U

node_regression_node_regr... U

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Avg Degree: 7.00

Max Degree: 9

rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)

\$ bash codeB/run.sh graphrnn graph_generation graph_generation

Running task: graph_generation using dataset: graph_generation

Loading saved model from: codeP/saved_models/graph_generation_graph_generation_graphrnn_graph_generation.pt

Generated Graphs:

Graph 1 → Nodes: 10 | Edges: 45

Avg Degree: 9.00

Max Degree: 9

Graph 2 → Nodes: 10 | Edges: 45

Avg Degree: 9.00

Max Degree: 9

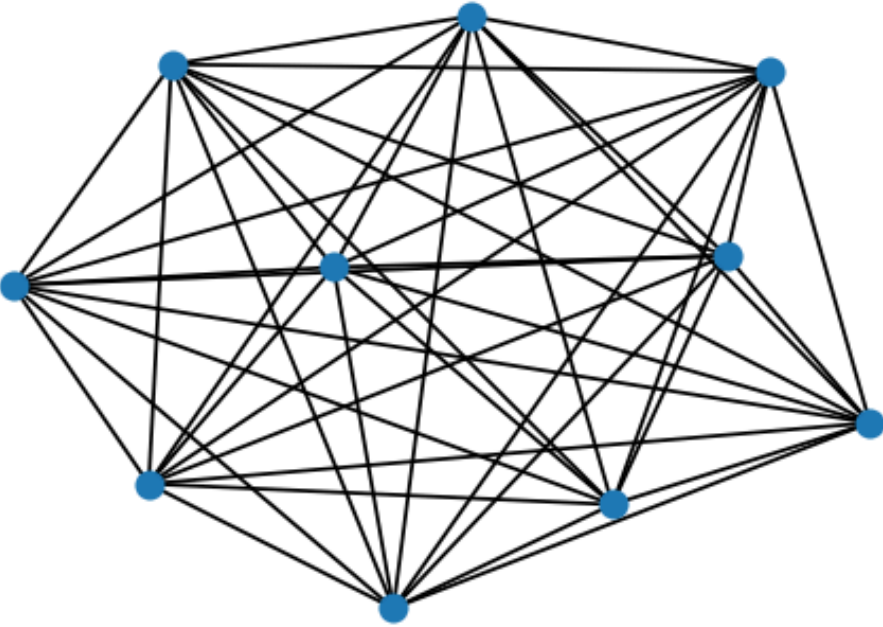
bash

bash

powershell

Figure 1

Generated Graph 2



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running graphrnn tasks for different tasks and datasets

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model_config.py

load_cora_edge.py

run.sh

dataset_config.py

codeP > model_config.py > ...

codeP > dataset_config.py > ...

83 "graphrnn": {

84 "module": {

PROBLEMS

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PORTS

Avg Degree: 7.00

Max Degree: 9

rahiiq@Rahiiq MINGW64 /c/Rahiiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiiq)

\$ bash codeB/run.sh graphrnn graph_generation graph_generation

Running task: graph_generation using dataset: graph_generation

Loading saved model from: codeP/saved_models/graph_generation_graph_generation_graphrnn_graph_generation.pt

Generated Graphs:

Graph 1 → Nodes: 10 | Edges: 45

Avg Degree: 9.00

Max Degree: 9

Graph 2 → Nodes: 10 | Edges: 45

Avg Degree: 9.00

Max Degree: 9

Graph 3 → Nodes: 10 | Edges: 45

Avg Degree: 9.00

Max Degree: 9

task_runner_files

dataset_config.py

edge_classification.py

edge_regression.py

example.py

gae.py

gat.py

gcn.py

gmn.py

grae.py

TLINE

ELINE

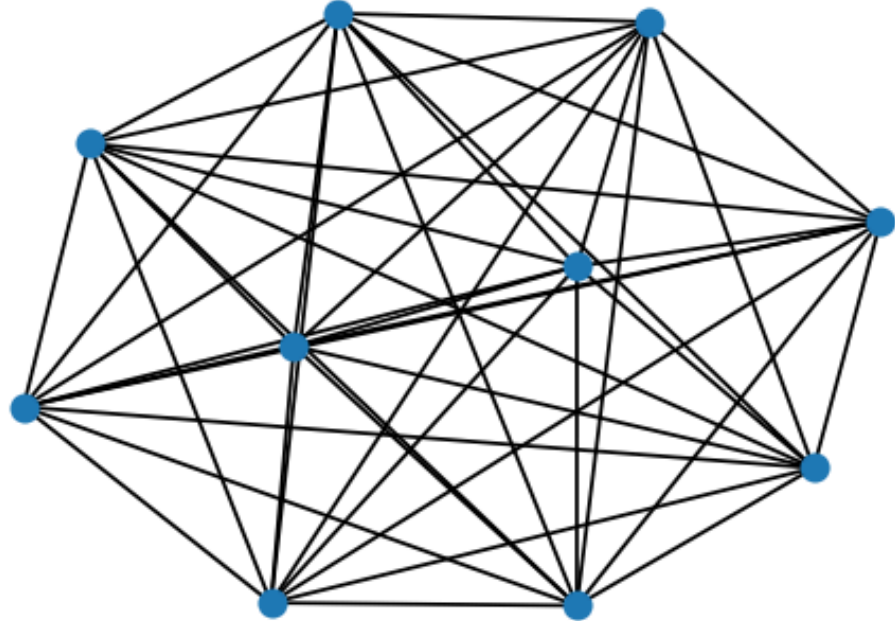
bash

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Figure 1

Generated Graph 3



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running grae tasks for different tasks and datasets

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh grae cora_reconstruction graph_reconstruction
Running task: graph_reconstruction using dataset: cora_reconstruction
[Match] Feature matrix from: x
[Match] Train edge_index from: train_edge_index
[Match] Test edge_index from: test_edge_index

Graph Statistics:
Number of nodes: 2708
Number of edges: 9500
Average degree: 7.02
Max degree: 167.0
No saved model found. Training from scratch...
[GRAE] Epoch 10/100 - Loss: 0.6987
[GRAE] Epoch 20/100 - Loss: 0.6947
[GRAE] Epoch 30/100 - Loss: 0.6936
[GRAE] Epoch 40/100 - Loss: 0.6934
[GRAE] Epoch 50/100 - Loss: 0.6932
[GRAE] Epoch 60/100 - Loss: 0.6932
[GRAE] Epoch 70/100 - Loss: 0.6932
[GRAE] Epoch 80/100 - Loss: 0.6932
[GRAE] Epoch 90/100 - Loss: 0.6931
[GRAE] Epoch 100/100 - Loss: 0.6931

Graph Reconstruction Results:
AUC: 0.5205
Average Precision (AP): 0.5208
F1-Score (threshold 0.5): 0.6665
Reconstruction Accuracy (threshold 0.5): 0.5038

```

Figure 1

ROC Curve

The figure is a plot titled "ROC Curve". The y-axis is labeled "True Positive Rate" and ranges from 0.0 to 1.0. The x-axis is labeled "False Positive Rate" and ranges from 0.0 to 1.0. A solid blue line represents the ROC curve, which is slightly above the dashed diagonal line (y=x). A text box in the upper left of the plot area states "AUC = 0.5205".

Navigation icons: Home, Back, Forward, Zoom In, Zoom Out, Full Screen, Save

Terminal sidebar with file explorer and task list

Task list: edge_classification, edge_regression, graph_classification, graph_generation, graph_matching, graph_reconstruction, graph_regression, link_prediction, node_classification, node_clustering, node_embedding

Footer: BLACKBOX Chat, Add Logs, CyberCoder, Improve Code, Share Code Link, Open Website, Generate Commit Message, CRLF, Python, Finish Setup, 3.11.5 (venv), Go Live, AI Code Chat, Prettier

running grae tasks for different tasks and datasets

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```
rahiq@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahiq)
$ bash codeB/run.sh grae cora_reconstruction graph_reconstruction
Running task: graph_reconstruction using dataset: cora_reconstruction
[Match] Feature matrix from: x
[Match] Train edge_index from: train_edge_index
[Match] Test edge_index from: test_edge_index

Graph Statistics:
Number of nodes: 2708
Number of edges: 9500
Average degree: 7.02
Max degree: 167.0
No saved model found. Training from scratch...
[GRAE] Epoch 10/100 - Loss: 0.6987
[GRAE] Epoch 20/100 - Loss: 0.6947
[GRAE] Epoch 30/100 - Loss: 0.6936
[GRAE] Epoch 40/100 - Loss: 0.6934
[GRAE] Epoch 50/100 - Loss: 0.6932
[GRAE] Epoch 60/100 - Loss: 0.6932
[GRAE] Epoch 70/100 - Loss: 0.6932
[GRAE] Epoch 80/100 - Loss: 0.6932
[GRAE] Epoch 90/100 - Loss: 0.6931
[GRAE] Epoch 100/100 - Loss: 0.6931

Graph Reconstruction Results:
AUC: 0.5205
Average Precision (AP): 0.5208
F1-Score (threshold 0.5): 0.6665
Reconstruction Accuracy (threshold 0.5): 0.5038

```

Figure 1

Precision-Recall Curve

The plot shows a Precision-Recall Curve. The x-axis is labeled 'Recall' and ranges from 0.0 to 1.0. The y-axis is labeled 'Precision' and ranges from 0.5 to 1.0. The curve starts at (0.0, 1.0), drops sharply to approximately (0.05, 0.87), and then follows a diagonal line down to (1.0, 0.5). A legend in the top right corner indicates 'AP = 0.5208'.

FILE EXPLORER

codeP

__pycache__

saved_models

edge_classification_edge_cl... U

edge_classification_edge_classifi... U

edge_classification_edge_cl... U

edge_regression_edge_regr... U

edge_regression_edge_regr... U

graph_classification_graph_... U

graph_classification_graph_... U

graph_classification_graph_... U

graph_generation_graph_genera... U

graph_matching_graph_mat... U

graph_matching_graph_mat... U

graph_reconstruction_grap... U

graph_reconstruction_grap... U

graph_regression_graph_re... U

graph_regression_graph_re... U

graph_regression_graph_re... U

link_prediction_link_predicti... U

link_prediction_link_predicti... U

link_prediction_link_predicti... U

node_classification_node_cl... U

node_classification_node_cl... U

node_classification_node_cl... U

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node_clustering_node_clust... U

node_clustering_node_clust... U

node_embedding_node_em... U

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```
rahik@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahik)
$ bash codeB/run.sh grae cora_reconstruction graph_reconstruction
Running task: graph_reconstruction using dataset: cora_reconstruction
[Match] Feature matrix from: x
[Match] Train edge_index from: train_edge_index
[Match] Test edge_index from: test_edge_index

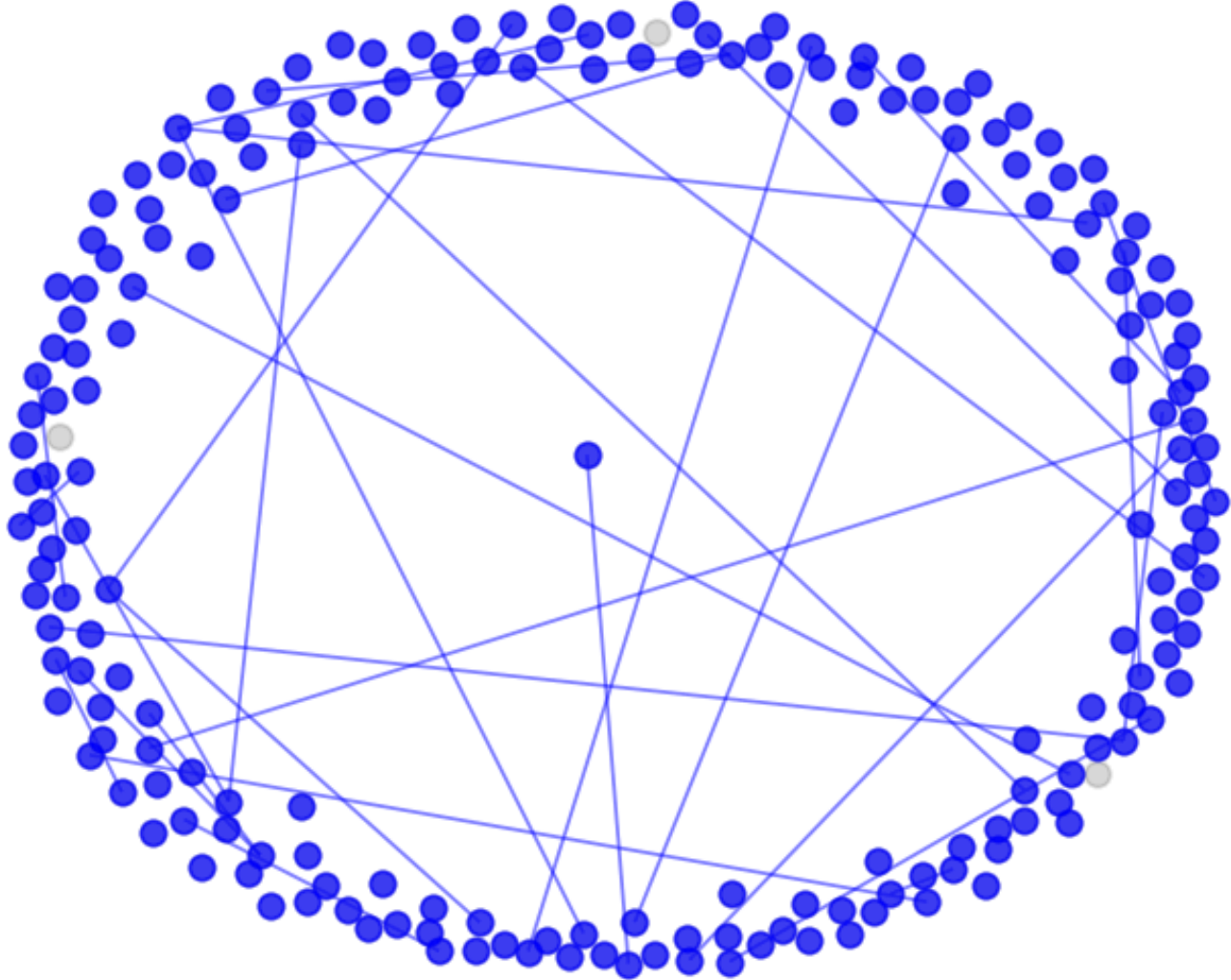
Graph Statistics:
Number of nodes: 2708
Number of edges: 9500
Average degree: 7.02
Max degree: 167.0
No saved model found. Training from scratch...
[GRAE] Epoch 10/100 - Loss: 0.6987
[GRAE] Epoch 20/100 - Loss: 0.6947
[GRAE] Epoch 30/100 - Loss: 0.6936
[GRAE] Epoch 40/100 - Loss: 0.6934
[GRAE] Epoch 50/100 - Loss: 0.6932
[GRAE] Epoch 60/100 - Loss: 0.6932
[GRAE] Epoch 70/100 - Loss: 0.6932
[GRAE] Epoch 80/100 - Loss: 0.6932
[GRAE] Epoch 90/100 - Loss: 0.6931
[GRAE] Epoch 100/100 - Loss: 0.6931

Graph Reconstruction Results:
AUC: 0.5205
Average Precision (AP): 0.5208
F1-Score (threshold 0.5): 0.6665
Reconstruction Accuracy (threshold 0.5): 0.5038

```

Figure 1

Original Graph Subgraph



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```
rahig@Rahiq MINGW64 /c/Rahiq/CSE299 Project/CSE299-Graph-Based--ML-Model (rahig)
$ bash codeB/run.sh grae cora_reconstruction graph_reconstruction
Running task: graph_reconstruction using dataset: cora_reconstruction
[Match] Feature matrix from: x
[Match] Train edge_index from: train_edge_index
[Match] Test edge_index from: test_edge_index

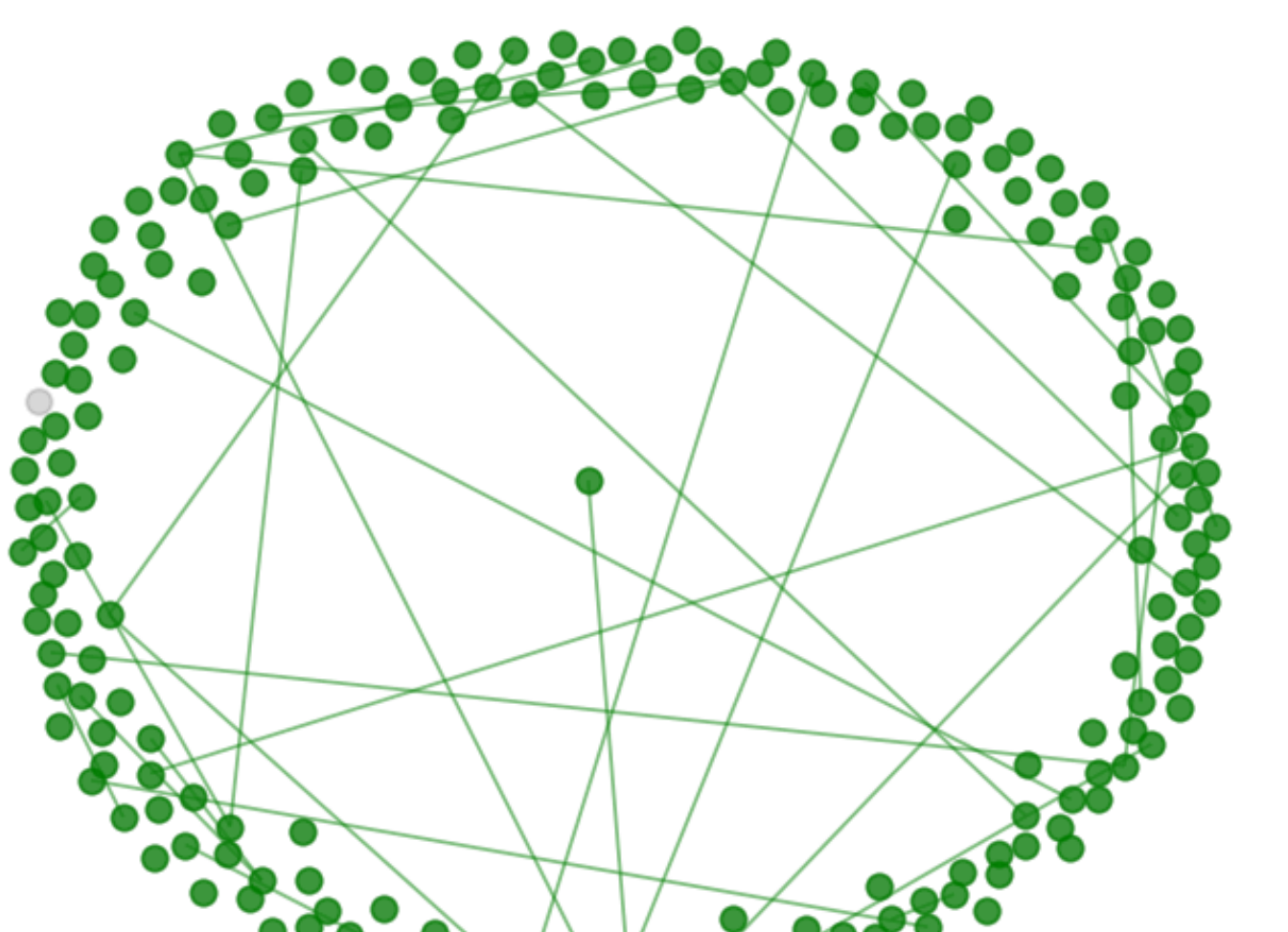
Graph Statistics:
Number of nodes: 2708
Number of edges: 9500
Average degree: 7.02
Max degree: 167.0
No saved model found. Training from scratch...
[GRAE] Epoch 10/100 - Loss: 0.6987
[GRAE] Epoch 20/100 - Loss: 0.6947
[GRAE] Epoch 30/100 - Loss: 0.6936
[GRAE] Epoch 40/100 - Loss: 0.6934
[GRAE] Epoch 50/100 - Loss: 0.6932
[GRAE] Epoch 60/100 - Loss: 0.6932
[GRAE] Epoch 70/100 - Loss: 0.6932
[GRAE] Epoch 80/100 - Loss: 0.6932
[GRAE] Epoch 90/100 - Loss: 0.6931
[GRAE] Epoch 100/100 - Loss: 0.6931

Graph Reconstruction Results:
AUC: 0.5205
Average Precision (AP): 0.5208
F1-Score (threshold 0.5): 0.6665
Reconstruction Accuracy (threshold 0.5): 0.5038

```

Figure 1

Reconstructed Graph Subgraph



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3.11.5 (venv)

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